M-INSTR spec: per-instrument profile layer (InstrumentProfile)

Rev 2.3, 2026-11-03 (ET). APPROVED for implementation: all questions decided (Q1–Q10, N1–N6). Rev 2.3 is final and ships with PR-M1b (#90 commits rev 2.2 byte-identical). Rev 2.2 was based on rev 2.1 as merged in `origin/main` @ `6853398` (PR #89, PR-M0; #88 merged as `cd7e50a`) and adds the PR-M0 dump results (§4.4.2). Data inputs still pending from Ettienne: Q3 IBKR per-side fees for the non-ES/MES roots and the Q7 bid/ask start dates; they do not block PR-M0 to PR-M3. Checked read-only against `github.com/ettiennehugo/MultiCharts.Net` `origin/main` @ `7dbdb1b` (merge of PR #85). PR #88 ("Land #86 + #87 on main") was OPEN at rev 2.1 and has since merged (`cd7e50a`); its code is quoted where marked "#88 head", and `_MtfMath.SymbolRoot` was at L108–119 (doc L108, method L109–119) on `6853398`, i.e. **main before #90**; after #90 (PR-M1a, head `a265f4d`) it is doc L108–119, method **L120–135**. Every file:line below was read on `7dbdb1b` unless marked **VERIFY** or "#88 head".

Prerequisite: PR #88 merges before any M-PR. #88 changes `_MtfMath` (around `SymbolRoot`, L29–39), `_CustomSignal` L1731–1830 (incl. the same-root check L1776–1785), `_DvpEngine`, the DVP golden fixtures and configHash. After it merges, `_CustomSignal` line numbers from about L1740 shift by up to about 50 lines; each M-PR re-reads its cites on the main it targets. Every M-PR targets `main` and is opened only after the previous one merged (no stacked PRs, §11).

Sources: the repo (every `*.CS`, `specs/*.md`, `tests/*`); `specs/instrument-profile-spec.md` rev 3 (2026-10-24, rev-3 Q1–Q11 DECIDED); `specs/m-dvp-spec.md` rev 6; `specs/m-port-spec.md` rev 2.7; the MultiCharts .NET Programming Guide, "Instrument properties" list (just before §4.2.4 StatusLine; box copy, not in the repo); Ettienne's typed API dump output `_McApiDump3.output.txt` (@ES, 2026-11-01; **box copy only, not in the repo**, and produced by a reflection dump, so it is cited as Ettienne's run output, not as a repo file; PR-M0 commits typed dump output as a fixture); CME contract-spec pages read 2026-11-02 (§8); the box research note `/workspace/m-instr/mc-sessions-research.md` (MC sessions: Programming Guide v1.1 §4.2.2–4.2.3, §4.7.4, MC wiki and forum staff answers, MC-1439; 2026-11-02); Ettienne's rulings (ES $17 and MES $2.50 per round trip, 2026-11-01; sessions from the chart, 2026-11-02; Q3–Q10 accepted, 2026-11-02).

Why a new file. Rev 3 treats "Future = today's ES behaviour" as an identity (IP-4.4) and classifies on `Category.ToString()` substrings (IP-1.1, IP-2.2). The 2026-11-01 audit found the futures path itself is ES-only (CME equity-index session clock, quarterly H/M/U/Z roll, third-Friday expiry, the root parser, the clusters). This spec **supersedes rev 3 where they conflict (§1.2)**; rev 3's DECIDED answers stay in force otherwise. Nothing from rev 3 I1–I6 is coded (`rg 'EInstrumentClass|InstrumentProfile' -g '*.CS'` finds nothing); rev-3 PR-I0 `_InstrProbe` is on main and is retired by PR-M1e.

Scope. M-INSTR is a **base service** in `_CustomSignal`. At `StartCalc()` (and, for session data, on the first `CalcBar` of each stream, §4.5) it: 1. reads every used stream's instrument through the **typed** API (`Bars.Info` / `BarsOfData(n).Info`, `IInstrumentSettings`) and the stream's **chart sessions** (`Bars.Sessions`, `IROList<SessionObject>`); 2. builds one immutable `InstrumentProfile` per stream from detection, the chart's sessions, a per-root **exchange-facts** preset (no times) and the user's **cost** figures; 3. logs detection and the applied configuration; 4. hands each engine its effective settings; 5. throws a clear `EngineNotSupportedException` where an engine cannot apply (one rule, R13).

Tuned parameters (thresholds, bandwidths, gap sizes, degrees) are **not** in profiles (R5).

Not touched. The PriceAction library. No new `[Input]`s. M-PORT behaviour (§9). ES on the current hosts and charts stays **bit-identical**, checked by the `_TestGate0` canary hash, the DVP D9a summary (post-#88 baseline) and the ES fixtures in `tests/*` (§10 MI-T15).

Revision log

Rev	Date (ET)	Change
1	2026-11-02	First draft (62-row audit, rev-3 conflicts C1–C8, Q1–Q10 OPEN).
2	2026-11-02	Coder review (37 items, Appendix A) applied, each re-verified on 7dbdb1b. **Sessions come from the chart (Q1/Q2 DECIDED by Ettienne):** no session/RTH windows in any preset; trading day, day open and session end derive from typed `Bars.Sessions`; RTH-window presets dropped. Q3–Q10 DECIDED (accepted recommendations). One throw-vs-degrade rule (R13). Lint scoped to MC enums with an allow-list. `_InstrProbe` retired in its own PR. Gate0Tests updated with the G0 tick change. ringBars kept literal. §9 aligned with the `willArm` code; cost floor is new opt-in behaviour. Audit 70 rows (8 added, 5 corrected, counts fixed). F1 widened, F3 second bug. `pointValue` → `bigPointValue`. M1 split four ways (+ probe retirement), M6 per engine. New questions N1–N6.
2.1	2026-11-03	Ettienne 2026-11-03 09:47 ET: "1 to 5 is fine, on 6 keep that dump in the same directory as the created standard log file". N1–N5 DECIDED as recommended; N6 DECIDED: `_InstrDump` writes `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt` into the directory of the host's standard `MCSignal*.csv` log, through the host's existing resolver (§4.4); no new hardcoded path. N4 becomes approved degraded mode D7. Status APPROVED for implementation.
2.2	2026-1	Base: rev 2.1 as merged in `origin/main` @ 6853398 (PR #89). **§4.3 root rule rewritten** (rev 2.1 steps 3–4 turned ESZ2024 into ESZ2024): leading digits + letter run, drop a final month letter followed by a digit when the run has ≥ 2 letters; MI-T2 gains ESZ2024, 'ES 12-24', ESZ24:XCME, CLZ2026, EUR/USD, BRK.B, Z4. **PR-M0 dump results** (Ettienne's 13 `_InstrDump` runs, §4.4.2): CONFIRMED `Bars.Sessions` filled at StartCalc and identical on the first bar (R12 now reads at StartCalc, re-checks on the first bar), session/bar times follow the chart time zone, per-series sessions, multiple sessions per day kept, `SessionName` value, 6E priceScale within the 1e-9 tolerance, SPY facts, CL/GC tick and big point value. **New MC**

Rev	Date (ET)	Change
	1-03	**behaviour:** on the GLOBEX template `LastBarInSession` did not fire at the 15:15 break, only at 16:00 → new **R14** (session ends from the Sessions list end times) and **MI-T16**. Still OPEN (awaiting Ettienne): Portfolio Trader second symbol not dumped; 6E Default session vs session-end bars (QuoteManager zone). B1/B2 not run (commented out in the build).
2.3	2026-11-03	Prepared after #90 (which commits rev 2.2 byte-identical); **final, ships with PR-M1b**. Header: the `SymbolRoot` L108–119 cite is main **before #90** (6853398); post-#90 location L120–135. §4.3/MI-T2: `MESH26` example replaced by `MESZ26/MESH27` → `MES`, as in the #90 tests. **@6E zone RESOLVED:** QuoteManager custom sessions with ET hours entered as CT (user config error, not MC behaviour); Ettienne switched to exchange sessions and will rerun; the 6E fixture is marked superseded. Folded in Coder's `mc-session-templates-research.md`: the built-in E-mini GLOBEX template is stale (2021 halt removal, pre-2012 15:30–16:30 row) and sits on the D9a/canary chart (environment caveat, engines take sessions as given); "Combined" = two sessions per day; templates editable in QuoteManager → Tools → Session Templates; `LastBarInSession` fires on rows with Session End ticked (inferred, not documented) as R14's rationale, with sources. Portfolio Trader @CL stays OPEN. **MC stores the root (credit: Ettienne):** `Bars.Info.ASymbolInfo2` (`MTPA_MCSymbolInfo2`, PG §4.2.3, pg11.txt L901, L3828–3867) becomes the M2 resolver's primary root, with the #90 parser as fallback when empty or flagged in `UnsupportedFields` (typed); `!SameRoot` → Warning, MC wins. §4.4 B1/B2 lead with `SymbolExpiry/SymbolCurrency`; new B3 cross-checks `SymbolCategory/MinMove/PriceScale/BigPointValue` vs `Bars.Info/IInstrument`. **Resolved by Ettienne `_RootDump` 2026-11-03, relayed by Coder (§4.4.3):** SymbolRoot/Currency/Category/Exchange/MinMove/PriceScale/BigPointValue CONFIRMED; parser = pure fallback on `(UnsupportedFields & eMTPA_SF_Root) != 0` or empty root; SymbolExpiry placeholder rule (B1); currency from SymbolCurrency (B2); MC-internal check PriceScale × Info.PriceScale = 1 ±1e-9 and BPV equal, else throw; MI-T2/MI-T3 fixtures; removed from §12.3. **@6E CLOSED:** rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` (17:00–16:00 Exchange, ends 16:00, matches ES; `bvo(TickCount)` so instrument/session facts only) is the 6E fixture; `01M6HSDCCJZXJHNB165KSF6Z5B.txt` discarded; 6E rerun removed from §12.3. Setup note: restart MC after changing QuoteManager sessions (cached).

0. Rules (normative; every PR must comply)

- **R1 Typed API only.** Never use MultiCharts enums or API members as strings: no `Enum.Parse/TryParse/GetNames` on an MC enum, no `<MC enum>.ToString()` used in a comparison, parse or hash, no reflection by member name on MC types. Formatting an MC enum into a human-readable log line is allowed (nothing reads it back). Repo-own enums parsed from protected config are outside R1 (§10 MI-T14 allow-list). Where a typed member is uncertain it is **VERIFY** with a typed dump line (§4.4); a line that does not compile is the answer (the member does not exist). Never guess.
- **R2 Build hygiene.** New `.CS` → `<Compile Include>` in `PLStudies.csproj` + one-time PowerLanguage Editor import (listed in the PR). New test projects → `PLStudies.Tests.sln`, `CheckForOverflowUnderflow=true`, real-source linking via a `gen.sh` awk region (`// <instr-core>` … `// </instr-core>`), as in `tests/InstrProbeTests`. C# 7.3.
- **R3 PriceAction unchanged.** The profile does not touch `EPriceActionPreset` or the PriceAction library; VPFUT keeps its existing preset call.
- **R4 No new `[Input]`s.** Settings go through protected config objects (`InstrumentConfig`, §5.3), like `EventConfig`, `RegimeConfig`, `DailyConfig`, `DvpConfig`, `PortMmConfig`.
- **R5 Self-calibration over tables.** Presets hold **exchange facts only**: tick, big point value, currency, contract months, expiry rule, roll rule, holiday calendar (where verified), cluster. **No session or RTH times** (Q1/Q2). ES-tuned values (e.g. `RollGapTicks` 8, `KdeBandwidthTicks` 6, `_ProfileStats` 4-tick disagreement) stay as today with a CFG note "ES-tuned default", or become relative to a running statistic in a later PR (§7).
- **R6 Auto-detect, apply, log, refuse.** No silent fallback. `TICK_FALLBACK = 0.01` (`_CustomSignal` L29, `resolveTick` L2287–2301 with a bare `catch {}`) becomes an Error + throw (PR-M1b).
- **R7 Volume basis.** Futures (`Future/FutureRolover`) on Build Volume On = Tick Count → Error + throw; non-futures on Tick Count are tagged `TickCount`. **Already on main** (`_VolumeProfileBase` L1429–1461, L1505–1506); M-INSTR only moves it into `volumeBasis`.
- **R8 Sizing.** 1% risk per trade; hard max-qty/max-notional caps reduce size (`sizeForRisk`, L2921–2953). The portfolio path (`_PortMM`, deny-not-reduce) is unchanged. Details and the actual code path: §9.
- **R9 Costs.** ES $17.00 and MES $2.50 per round trip (Ettienne). Other roots: method DECIDED (Q3), figures need Ettienne's IBKR per-side fees per root (data input).
- **R10 Logs.** One `File.AppendAllText` per line; skipped entries at **Info**; doubles formatted with `CultureInfo.InvariantCulture`.
- **R11 Facts.** Cite file:line; exchange facts carry a source or **VERIFY**.
- **R12 Sessions come from the chart (DECIDED, Q1/Q2).** Session times, the trading-day boundary and the session end come from the stream's chart sessions via typed `Bars.Sessions` / `SessionObject`. The chart's session template is the source of truth (normally the instrument's MC "Default"). Engines do not second-guess it and do not warn about which template the user chose. Sessions are per stream (each data series has its own, **CONFIRMED** §4.4.2; in Portfolio Trader each symbol has its own, with one global time zone) and are read **in StartCalc and re-checked on the first CalcBar** of each stream (both reads identical in all 13 dump runs, **CONFIRMED** §4.4.2; a difference throws `sessions:changed`, S1). `Bars.Sessions` exists (it compiles on main) but is documented only on the MC forum; its behaviour is now taken from the PR-M0 dump (§4.4.2), not from the documentation. The template name (`Info.SessionName`, Programming Guide) is logged and never parsed. Holidays cannot be read from code, so holiday calendars stay in the repo as calendars only (PR-M8). The DV/MTF session-mismatch Warning between streams (#86/#88 `checkHtfSessions`), including its MC-1439 / Realtime-History Matching caveat, stays.
- **R13 Throw vs degrade (one rule).** An engine **throws** when it cannot apply to the instrument or to the chart's sessions. It runs **Degraded** only in a mode Ettienne has approved, which still produces correct results from reduced inputs; each such mode is listed in §6.4 with what is reduced. An output that is undefined for the given sessions (e.g. an overnight high on a chart with no overnight segment) is NaN, logged once at Info, and is not a degrade; §6.4 lists every such output.
- **R14 Session ends come from the Sessions list (NEW, rev 2.2).** A session ends at the `EndTime` of a `SessionObject` in the stream's Sessions list, not where `Bars.LastBarInSession` happens to be true. Observed (§4.4.2, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`): on the CME GLOBEX template (sessions 17:00–15:15 and 15:30–16:30 CT) `LastBarInSession` was true only on the 16:00 bar, **not at the 15:15 break**. The session reader and every session-reset engine (VPFUT, AVWAP, FLOW, TOD, RANGE, WEEKLY, DAILY, DVP, G0, REGGATE, ROLL keying) therefore assign each bar to the session containing its close (start exclusive, end inclusive) and detect a session end as: (a) a bar whose close equals the session's `EndTime`, or (b) the first bar that falls in a different session than the previous bar (the end is then attributed to the previous bar), or (c) `LastBarInSession` true (kept only as an extra early signal for the last bar when data stops before the listed end, e.g. ES data ends 16:00 inside a session listed to 16:30). `LastBarInSession` alone is never the only signal. A reset fires **once per (stream, trading day, session index)**: the two-series run logged the same 2026-10-26 16:00 Data1 bar as a session end twice (`01M6HSDBFH2D6H21PMYXYKFJHV.txt`, CalcBar called once per stream update), so resets must be idempotent. Tested by MI-T16. **Rationale (rev 2.3):** each template row has a **Session End** checkbox and MC staff say to tick it on the row that ends the trading day (forum t=45178, https://www.multicharts.com/discussion/viewtopic.php?t=45178); ticking it on every row gave a break at every row (t=5890, https://www.multicharts.com/discussion/viewtopic.php?t=5890). `LastBarInSession` therefore most likely fires only at rows with Session End ticked, which matches the GLOBEX observation (15:15 row presumably unticked). This is **inferred, not documented** (research `/workspace/m-instr/mc-session-templates-research.md` §6), and the tick is not exposed by `SessionObject` (§4.1), so the list `EndTimes` are the only readable boundary; live, the flag can also be late or missed when no tick prints in the last bar (staff, https://www.multicharts.com/discussion/viewtopic.php?t=50358; https://www.multicharts.com/discussion/viewtopic.php?t=48918).

1. Relation to `instrument-profile-spec.md` rev 3

1.1 Kept from rev 3

The engine × class support matrix (rev 3 §8.1) and its DECIDED answers: rev-3 Q1 ETF = Stock; rev-3 Q2 FLOW throws on stocks; rev-3 Q3 VPFUT throws on non-futures; rev-3 Q4 no force flag; rev-3 Q5 DVP/G0 Unsupported on FX, VP Degraded (visual) on FX; rev-3 Q7 `profileHash` outside configHash; rev-3 Q8 `_RegimeEngine` on ES/MES only; rev-3 Q9 ES $17; rev-3 Q10 USD only; rev-3 Q11 EVENT Degraded on USD FX pairs. Also kept: the INSTR/CFG log idea, the `EngineNotSupportedException` format (rev 3 §6), the cost-floor formula (IP-8.3).

1.2 Conflicts (this spec wins)

#	Rev 3	Conflict	Rev 2
C1	IP-1.1/IP-2.2 classify by `Category.ToString()`; probe uses `Enum.GetNames`	R1	Exhaustive `switch` on `ESymbolCategory` in the host (§4.2).
C2	IP-1.4 "Build Volume On is not on `IInstrumentSettings`"	Wrong: `Info.BV0` is typed (`_VolumeProfileBase` L1505)	`volumeBasis` from `Info.BV0`.
C3	IP-4.4 "Future = today identity"	Futures path is ES-only	Per-root exchange facts (§8) + chart sessions; ES bit-identical.
C4	IP-8.1 "q is never reduced"	Hard caps reduce today (L2943–2949)	§9 describes the actual code.
C5	IP-8.3 MES $2.47	R9	MES $2.50, floor 20 ticks (integer-cents formula, §9).
C6	PR-I0 `_InstrProbe` uses reflection (`_InstrProbe` L4/67/74/75/144; `_InstrProbeCore` L5/54/126/167)	R1	Typed `_InstrDump` (PR-M0); `_InstrProbe` + `tests/InstrProbeTests` retired in PR-M1e (Q6 DECIDED).
C7	Rev-3 line numbers	Stale	Re-read on `7dbdb1b`.
C8	PR-I4 "no futures holiday re-route"	Other CME product groups have other holidays/early closes	Calendar rule §6.4 S4 and PR-M8; ES keeps `_CmeHolidays`.
C9	Rev-3 session handling = CME equity-index clock for every future	R12	Sessions from `Bars.Sessions` (§4.5); no session times in presets.

2. Purpose

Today an engine run on anything other than @ES/@MES silently uses ES facts, guesses from strings, or fails later with an unclear error. M-INSTR: (1) detects the instrument from typed MC data; (2) takes session times, trading day and session end from the chart's sessions; (3) resolves the exchange facts each engine needs (tick, big point value, currency, contract months, expiry and roll rule, calendar, cluster) from detection plus a small exchange-facts preset per root; (4) applies them, ES bit-identical; (5) logs; (6) refuses clearly under R13.

3. Audit: ES- and futures-specific hardcoding on `origin/main` @ `7dbdb1b`

Method: `rg` over every `*.CS`, `specs/*.md`, `tests/*` for 0.25, 12.5, $50, `"ES"`/`@ES`, H/M/U/Z, `ThirdFriday`, `TimeSpan(9,30)`, (16,0), (16,15), (17,0), (18,0), (15,15), Globex, Eastern/Central, USD, `Category.ToString`, `SymbolRoot`, `PortfolioCluster`, cost literals (471 raw hits, `/workspace/m-instr/grep1.txt`), reduced to one row per distinct hardcoding, plus the 8 rows the rev-1 review found missing (63–70). Column "Sess" = **S** where the row is replaced by chart-session data under R12 (§4.5), **T** where it is a frozen pre-registered trial value (kept, coverage-checked, §6.4 S3).

Engine tags: CS host `_CustomSignal`; VPFUT `_VolumeProfileFuturesBase`; CLOCK `_CmeSessionClock`; CAL holiday calendars; ROLL `_RollDetector` + host roll code; EVENT `_EventCalendar`; G0 `_Gate0`/`_TestGate0`; FLOW `_OrderFlow`; REGGATE `_RegimeGate`/`_Test6`; REGIME `_RegimeEngine`; DAILY `_DailyContext`; TOD `_TodNormaliser` (host side); DVP `_DvpEngine`/`_DvpEngineTest`; SIZE CS-4 sizing; COST; PORT M-PORT; MTF `_MtfMath`; LEVEL, AVWAP, RANGE, WEEKLY, PSTATS `_ProfileStats`; LEGACY pre-engine studies.

3.1 Table

#	file:line	What is hardcoded	Engine	Profile field	Sess
1	`_CustomSignal` L1603–1618	`PortfolioCluster` switch on root strings: EQIDX (L1611), RATES (L1612), ENERGY (L1613), METALS (L1614); others fall through to the raw root	PORT	`cluster`	

#	file:line	What is hardcoded	Engine	Profile field	Sess
2	_CustomSignal L2462	autoTradingDay(barTime, Bars.Info.Category.ToString()): MC enum → string (R1)	CS	tradingDay (typed)	S
3	_CustomSignal L2581–2587	autoTradingDay substrings "FOREX"/"CURRENCY"/"CRYPTO": FX day = ET+7h; crypto = UTC date	CS	tradingDay	S
4	_CustomSignal L2614–2626	exchangeTimeZoneId substring map: L2619 CME/CBOT/CBT/GLOBEX/CBOE→Central; **L2621 NYMEX/COMEX/NYSE/NASDAQ/ARCA→Eastern** (CME Group quotes CL/GC hours in CT; dump: @CL shows 18:00–17:00 on both Exchange and Local, @GC 18:00–17:00 on Exchange, i.e. MC's exchange zone for NYMEX/COMEX on Ettienne's setup equals ET, §4.4.2 A3; whether that is the QuoteManager exchange zone or a template entered in ET is not separable from these runs, so the map entry stays as is and is not "fixed" to Central); other exchanges throw (L2603)	CS	barZone (only for ET conversions, §4.5)	S (partly)
5	_CustomSignal L29, L2287–2301	TICK_FALLBACK = 0.01 with a bare catch {}	CS	tick (throw)	
6	_CustomSignal L2062	_NyseHolidays created for every instrument	CAL	calendar	
7	_CustomSignal L3228–3240	tradingDayCount uses NYSE for non-stocks (warns once)	CAL	calendar	
8	_CustomSignal L921	isRthBar returns true on the generic base	CS	rthSplit (§4.5, N1)	S
9	_CustomSignal L949	todMinuteOfSession default; futures override = 18:00 ET (row 18)	TOD	dayOpen	S
10	_CustomSignal L1063–1072	DAILY roll-offset warning assumes the 18:00 ET splice; Category == Future only (misses FutureRolover, F2)	DAILY	tradingDay, assetClass	S
11	_CustomSignal L1801, L1836	RollGapTicks = 8 (ES-tuned; _RollDetector L12–13)	ROLL	none (Q4)	
12	_CustomSignal L1866, L1881–1882	Roll warning / m_rollGlobexKey require Category == Future; isFuturesCategory (_VolumeProfileBase L1440–1442) also accepts FutureRolover	ROLL	assetClass	
13	_CustomSignal L1840–1842	Roll inheritance keyed by _MtfMath.SymbolRoot (row 31). (Rev 1 cited L1778–1783, which is the HtfStreams same-root throw L1776–1785, also a SymbolRoot user, fixed by PR-M1a.)	ROLL	root	
14	_CustomSignal L749, L766	DVP root = SymbolRoot(...).ToUpperInvariant()	DVP	root	
15	_CustomSignal L2891–2953	unitNotional 0 = entry × BigPointValue, implicit USD	SIZE	bigPointValue, currency	
16	_CustomSignal L2994–3002	roundTripCostPerContract = 2 × (MC Commission + Slippage)	COST	cost.roundTrip	
17	_CustomSignal L2008	AVWAP sessionDate = tradingDayKey, rth = isRthBar	AVWAP	tradingDay, rthSplit	S
18	_VolumeProfileFuturesBase L489–493	TOD minute = minutes since 18:00 ET	TOD	dayOpen	S
19	_VolumeProfileFuturesBase L52–63	DEFAULT_* strings in ET: Asia, Europe, NY, RTH 09:30–16:15, halt 17:00–18:00, Globex 18:00/17:00	VPFUT	session data / N1 / N3	S
20	_VolumeProfileFuturesBase L66–77	Same defaults as TimeSpans (VPF-13)	VPFUT	as row 19	S
21	_VolumeProfileFuturesBase L361	tradingDayKey = GlobexDate (18:00 ET)	VPFUT	tradingDay	S
22	_VolumeProfileFuturesBase L396–398	DefaultRthValueEnd 16:00 ET (value window, D5)	VPFUT	rthSplit (N1)	S
23	_VolumeProfileFuturesBase L472–476	EOwnProfileWindow { MultiChartsProfile = 0, GlobexDay, Rth }	VPFUT	names stay; GlobexDay = chart trading day	S
24	_VolumeProfileFuturesBase L517; _CmeSessionClock L373–379	isRthBar = RTH **value** window (09:30, 16:00] ET via IsRthValueBarEt; m_rthEnd 16:15 is the session flag, not the value window	VPFUT	rthSplit (N1)	S
25	_VolumeProfileFuturesBase L975–995	Tick-chunk plan per Globex day	FLOW	tradingDay	S
26	_VolumeProfileFuturesBase L1627, L1635–1671	VPFUT's own UseCmeHolidayTable (L1627) and hooks: IsGlobexClosedDay L1635, IsNoRthDay via NYSE L1641–1646, HaltEt L1656, RthCloseEt L1665, _CmeHolidays.Rules.Get(date).Closed L1671	CAL	calendar	
	_CmeSessionClock L48–61	"CME equity-index hours" defaults in ET: RTH, RthValueEnd, halt, Globex open/close, Asia/EU/NY			S

#	file:line	What is hardcoded	Engine	Profile field	Sess
27			CLOCK	session data / N1 / N3	
28	_CmeSessionClock L81	Clock's UseCmeHolidayTable = true (separate from VPFUT's L1627, row 26)	CAL	calendar	
29	_CmeSessionClock L248–249	GlobexTradingDayEt(et) 18:00 ET open	CLOCK	tradingDay	S
30	_CmeHolidays L13–45, L65	CME equity-index holiday/early-close rules (13:15 ET L44, Good Friday 09:15 ET L45, halts L65)	CAL	calendar (stays; not session times)	
31	_MtfMath L29–39	SymbolRoot: letters only; drops a trailing month letter. "@6E", "6EZ26" → ""; "EUR/USD" → "EUR" (F1)	MTF	root	
32	_RollDetector L23	gapTicks = 8	ROLL	none (Q4)	
33	_RollDetector L43	RollRuleMondayFrom = 2022-04-01	ROLL	roll	
34	_RollDetector L46–51, L56–67	ThirdFriday, quarterly SeriesWindow	ROLL	contractMonths, expiry, roll	
35	_EventCalendar L20	Event rows in ET (correct for US macro)	EVENT	barZone for conversion	
36	_EventCalendar L128	Default event classes (no EIA for ES)	EVENT	eventClasses	
37	_EventCalendar L189	EventRules.RollRuleMondayFrom 2022-04-01	EVENT	roll	
38	_EventCalendar L290–303	ExpiryDate/RuleExpiryDate 3rd Fri; RollDate Mon before (2022+) else Thu −8	EVENT	expiry, roll	
39	_EventCalendar L1913–1928	nextQuarterlyExpiry, SessionsToExpiry, RuleRollDate: quarterly	EVENT	contractMonths	
40	_Gate0 L328	KdeBandwidthTicks = 6	G0	none (ES canary)	
41	_Gate0 L357	tick = 0.25 default (host sets it: _TestGate0 L237; _Gate0 L461 study.tickSize = tick)	G0	tick	
42	_Gate0 L474, L614, L1037/1059/1116/1129/1267, L1070	RTH 09:30–16:00 ET, 30-min periods, flat 15:55, 09:35	G0	frozen trial windows	T
43	_Gate0 L1185–1186	Doc: "q × widened stop × $50"; costFloorTicks only in comments	G0	doc	
44	_Gate0 L1204	new _DvpEngine(c, "G0KD02", "ES", tick, …)	G0/DVP	root	
45	_TestGate0 L42, L156–158	"CME Globex" template; $2.25/side + 0.5 tick/side = $17	G0	cost	
46	_OrderFlow L124	TickChunkBoundaryEt = 17:00	FLOW	sessionEnd	S
47	_OrderFlow L935	PlanGlobexDays(...)	FLOW	tradingDay	S
48	_RegimeGate L45–48	A2-63: rthOpen 09:30, activeEnd 15:30, exit 15:55, rthClose 16:00 ET	REGGATE	frozen trial windows	T
49	_RegimeGate L55–62	Roll years, digest fdd63bd8f179a8d4 (L60), tradingDay = "Globex18ET" (string, L62)	REGGATE	roll, typed key	
50	_Test6 L63, L89, L105–106	ET zone string, Globex key, RollDetection.ToString() (repo enum) into the hash		typed key	S

#	file:line	What is hardcoded	Engine	Profile field	Sess
			REGGATE		
51	_DailyContext L30, L33	{252,126,63}, hvAnnualisation 252	DAILY	sessionsPerYear	
52	_RegimeEngine L272	sqrt(63/252)	REGIME	sessionsPerYear	
53	_LevelStudy L79	tickSize = 0.25 (host sets L1410)	LEVEL	tick	
54	_AvwapManager L153	tickSize = 0.25 (host sets L1976; L195 rejects ≤ 0)	AVWAP	tick	
55	_RangeProfile L31	ringBars = 2880 (a **bar** count)	RANGE	none: literal kept	
56	_WeeklyProfiles L19, L26	Default source RthSessions	WEEKLY	rthSplit (N1)	S
57	_PortMM L388, L740; _PortfolioRiskManager L87, L129	USD-only book (typed eMTPA_MCSC_USD, correct)	PORT	currency	
58	_DvpEngineTest L213, L230	root == "ES": roll table ES-only	DVP	roll	
59	_DeltaDivRevFuturesLELX L152, L175, L528–564	Globex state, Globex-day trade limit	LEGACY	note only	
60	_ExitHalfRiskHalfRoll L107–113	"One tick" = 1 / PriceScale * BigPointValue (0.5 on ES if PriceScale = 100: neither a tick nor $/tick) (F3a)	LEGACY	tick	
61	_EconomicCalendar L231; _EhlersFilter L149; _StructuralTargetPatterns.Strategy L1060; _GenerateNNTrainingData L354	FX pip = PointValue × 10 × BigPointValue	LEGACY	note only	
62	_EpisodicPivotBase L32–35; _ParabolicShortSESX L51–52; _BreakoutAutoLELX L42; _TrendLine L2755–2756; _VolumeProfileBaseLog L206, L426	US-stock 09:30/16:00 ET; m_tick = 0.25	LEGACY	note only	
63	_CustomSignal L1888–1889	findTz("Eastern Standard Time") + GlobexTradingDayEt for the roll key on any future	CS	tradingDay (removed)	S
64	_DvpEngineTest L119–121; _Test6 L100–102	Host-generated quarterly RollDates (for m = 3..12 step 3: RollDates.Add(EventRules.RollDate))	DVP	contractMonths, roll	
65	_ProfileStats L47	ringBars = 1440 (bar count)	PSTATS	none: literal kept	
66	_ProfileStats L575	disagree when	mcPoc – cvPoc	> 4 × tick (ES-tuned)	PSTATS
67	_TestGate0 L157	NQ slippage $2.50 in the canary note	G0	cost (doc)	
68	_MtfMath L28	SymbolRoot doc says returns "@ES"; code returns "ES"	MTF	doc fix	
69	_DvpEngine L2445–2447	RootError: empty == empty is a match, so any two digit-led roots pass (F1)	DVP	root	
70	_ExitHalfRiskHalfRoll L123	GenerateStopLoss(m_firstStopPrice): a price distance passed where MC takes currency (F3b)	LEGACY	bigPointValue	

Counts. 70 rows; each counted in exactly one area. Rows 61 and 62 group several legacy files each.

Area	Rows	Count
CS host	2, 3, 4, 5, 8, 9, 10, 63	8
VPFUT	18–24	7
CLOCK and CAL	6, 7, 26–30	7
ROLL	11, 12, 13, 32–34	6
EVENT	35–39	5
G0	40–45, 67	7
FLOW	25, 46, 47	3
REGGATE	48–50	3
DAILY and REGIME	51, 52	2
SIZE and COST	15, 16	2
PORT	1, 57	2
DVP	14, 58, 64, 69	4
MTF	31, 68	2
AVWAP	17, 54	2
LEVEL, RANGE, WEEKLY, PSTATS	53, 55, 56, 65, 66	5
LEGACY	59–62, 70	5
Total		**70**

Tests. ES literals by project: VpHelper 102, OrderFlow 60, Gate0 45, Roll 39, PortMM 23, Dvp 16, Gmm 15, PriceAction 10, fewer elsewhere. These are ES fixtures by design and stay as the bit-identity guard; M-INSTR adds per-root fixtures. Exception: Gate0Tests is edited in PR-M6a (tick, §7).

3.2 Bugs today (fixed independently, PR-M1a–c)

- **F1 (wider than rev 1).** `SymbolRoot` drops digit-led roots ("@6E", "6EZ26" → "") and truncates FX pairs ("EUR/USD" → "EUR", colliding across EUR pairs). `_DvpEngine.RootError` L2445–2447 then treats "" == "" as a match, so any two digit-led roots pass the same-root check, and roll inheritance (L1840–1842) and the HtfStreams check (L1776–1785) are wrong for them. Fix in PR-M1a (§4.3); `RootError` treats an empty root as an error.
- **F2.** `Category == ESymbolCategory.Future` at L1069, L1866, L1882 misses `FutureRolover`; use the host's `isFuturesCategory` (`_VolumeProfileBase` L1440–1442). PR-M1b.
- **F3 (two bugs).** (a) `_ExitHalfRiskHalfRoll` L107–113 computes `1/PriceScale*BigPointValue`, which is neither a tick nor $/tick; (b) L123 passes `m_firstStopPrice`, a price distance, to `GenerateStopLoss`, which takes a currency amount. Fix: tick = `MinMove/PriceScale`; amount = distance × `BigPointValue` (per contract, matching the strategy's existing per-contract/position mode, VERIFY in the PR). PR-M1c.
- **F4.** `TICK_FALLBACK = 0.01` (L29) with a bare `catch {}` in `resolveTick` (L2287–2301) → Error + throw. PR-M1b.

3.3 MC-enum string uses on main (findings for PR-M1b / allow-list for MI-T14)

file:line	Use	R1 status	Action
`_CustomSignal` L2462	`Category.ToString()` passed to `autoTradingDay` substring tests	violation	PR-M1b: typed `switch (ClassOf)`
`_InstrProbe` L67, L74, L75	`Category.ToString()` / reflection	violation	PR-M1e: retire the probe
`_CustomSignal` L279	`TimeZone.ToString()` into a log line	allowed (log)	none
`_CustomSignal` L1883	`res.ToString()` (`EResolution`) into a Warning text	allowed (log)	none
`_TestPortMM` L90	`StrategyCurrencyCode.ToString()` log	allowed (log)	none
`_Test6` L105	`RollDetection.ToString()` (repo enum `ERollDetection`, `_CustomSignal` L1791) into the A2-63 hash	outside R1 (repo enum) but hash-fragile	PR-M6h: explicit switch, same spellings

4. Detection (typed MC API)

4.1 Members used

Status: **confirmed-PG** = in the MultiCharts .NET Programming Guide (v1.1 §4.2.3 "Properties of the Data Series", pp. 21–22; `mc-sessions-research.md` §6); **compiles** = used in code on main (so the member exists); **VERIFY** = neither, or a behaviour (not the member) is unconfirmed.

Member	Type	Status	Source
`Info.Category`	`ESymbolCategory`	compiles; values Future 0, FutureOption 1, Stock 2, StockOption 3, Index 4, CurrOption 5, MutualFund 6, MoneyMarketFund 7, IndexOption 8, Cash 9, Bond 10, Spread 11, Forex 12, CPCSymbol 13, Composites 14, FutureRoot 15, FutureOptionRoot 16, StockOptionRoot 17, FutureRolover 18, Commodity 19, Crypto 20, Warrant 21, None 254 from Ettienne's dump3 run (box copy, not in repo)	`_VolumeProfileBase` L1442, `_CustomSignal` L1069
`Info.MinMove`, `Info.PriceScale`	double	compiles, PG	`_CustomSignal` L2291–2292
`Info.BigPointValue`	double	compiles, PG ("in the instrument currency")	`_CustomSignal` L1660, L1678, L2360
`Info.PointValue`	double	compiles, PG (a **different** member from BigPointValue)	`_EconomicCalendar` L231
`Info.Description`	string	compiles, PG	`_EconomicCalendar` L243, L679
`Info.Exchange`	string (data value; parsing allowed)	compiles, PG	`_CustomSignal` L2601–2603
`Info.Name`	string (data value)	compiles, PG	`_CustomSignal` L1886
`Info.Resolution.Type` / `.Size`	`EResolution` / int	compiles, PG	`_CustomSignal` L1069, L966, L2245
`Info.TimeZone`	`RequestTimeZone` {Local, Exchange, GMT}	compiles, PG	`_CustomSignal` L279, L2594–2606
`Info.BVO`	`BuildVolumeOn`	compiles	`_VolumeProfileBase` L1505
`Info.SessionName`	string	**confirmed-PG**; log only, never parsed. Returns the template name as set on the chart: `Default` for Default, `CME: Equity Index Futures [E-mini] (GLOBEX)` for the GLOBEX template (**CONFIRMED** `01M6HSDCPGEWFS9VAYZG2C808V.txt`, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`)	PG §4.2.3, dump
`Info.Request`	`DataRequest`	confirmed-PG; not used by M-INSTR	PG §4.2.3
`Bars.LastBarInSession`	bool	confirmed-PG (§4.2.2). **Not a reliable session-end signal on its own**: not true at the GLOBEX 15:15 break (`01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`); see R14	PG, dump
`Bars.Sessions`	`IROList<SessionObject>` (`.Count`, `[i]`)	**compiles** (`_CheckData` L54 in StartCalc, L198–200; `_QullamaggieBase` L189–192; #88 head `mtfSessions` via `IInstrument`), so the members exist; **not in the PG** (forum/staff only). Populated at StartCalc and identical on the first CalcBar: **CONFIRMED** in all 13 runs (`sameAsStartCalc(yes)`, §4.4.2)	main, #88 head, dump
`SessionObject.StartDay/EndDay` (DayOfWeek), `.StartTime/.EndTime` (TimeSpan)		compiles (same sites); zone of the times = the chart (Format Instrument) time zone of the stream: **CONFIRMED** (@ES 17:00–16:00 on Exchange `01M6HSDCPGEWFS9VAYZG2C808V.txt`, 18:00–17:00 on Local `01M6HSDBFJ9WRF71G51B82VKCH.txt`; bar times shift with it)	main, dump
`SessionObject.EndOfSession(DateTime)`, `Sessions.SessionForDate(DateTime)`, an end-of-session bool		forum only: **VERIFY**; M-INSTR does **not** use them	research §6

Member	Type	Status	Source
`Info.ASymbolInfo2`	`MTPA_MCSymbolInfo2` (COM struct)	**confirmed-PG**: listed under §4.2.3 "Instrument properties" (`MTPA_MCSymbolInfo2 ASymbolInfo2 { get; }`, pdftotext `/workspace/research/pg11.txt` L901); struct fields in the `MTPA_MCSymbolInfo2` section (L3828–3867, PG p. 82): `SymbolName`, **`SymbolRoot`** ("symbol root (for futures), for example ES, CL", L3833), `SymbolDescription`, `SymbolExchange`, `SymbolCategory` (`MTPA_MCSymbolCategories`), **`SymbolCurrency`** (`MTPA_MCSymbolCurrency`), **`SymbolExpiry`** (`DateTime`, "expiration date (for futures)"), `MinMove` (int), `PriceScale` (double; PG example "for ES = 1/100", L3853, whereas `Info.PriceScale` dumped `100`), `BigPointValue`, `PutOrCall`, `StrikePrice`, **`UnsupportedFields`** (`MTPA_SymbolFields`, "fields which are not provided by the source"); "received through … `Bars.Info.ASymbolInfo2` for the current symbol" (L3867). **CONFIRMED** (Ettienne `_RootDump` 2026-11-03, relayed by Coder, dxFeed; @ES, @CL, @6E, SPY, table in §4.4.3): `SymbolRoot`, `SymbolCurrency`, `SymbolCategory`, `SymbolExchange`, `MinMove`, `PriceScale` (= 1/`Info.PriceScale`: 0.01 vs 100; 6E 1E-05 vs 99999.999999999985) and `BigPointValue` (= `Info.BigPointValue`); `SymbolName` = `Info.Name` in all four; `UnsupportedFields` = 0 in all four. `SymbolExpiry` is a **placeholder** on continuous futures (2038-01-01) and stocks (1900-01-01) (§4.4 B1). `MTPA_SymbolFields` is **not** `[Flags]` but its values are powers of two: eMTPA_SF_Root=1, Exchange=4, Description=8, Category=16, Expiry=32, PutOrCall=64, StrikePrice=128, Multiplier=256, Currency=512, ALL=-1. Credit: Ettienne, rev 2.3	PG §4.2.3, PG p. 82
`Info.Expiration`	`DateTime`	confirmed-PG as a property; never compiled in the repo (the probe reads it by reflection): **VERIFY**, and **not needed**: expiry comes from `SymbolExpiry` on dated contracts, else the roll calendar/ detector (§4.4 B1); not used by M-INSTR	PG §4.2.3
`StrategyCurrencyCode`	`MTPA_MCSymbolCurrency`	compiles on an `IStrategy` from a portfolio signal only (`_PortfolioRiskManager` L125–129, `_TestPortMM` L90); on a plain `SignalObject`: **VERIFY**, and **not needed**: currency comes from `ASymbolInfo2.SymbolCurrency` (CONFIRMED, §4.4 B2); not used by M-INSTR	main
Per-symbol currency on `Info`	—	no documented member; none guessed. Currency comes from the preset (USD, Q8)	—
Exchange time-zone member, `SessionStartBar`/ `SessionEndBar`, "is Default" flag	—	not documented; not used	research §6
Holidays from code	—	**not available** (MC staff, 2013/2015/2022)	research §6

4.2 Classification (host, exhaustive switch; replaces rev-1 MI-T1)

```
static EAssetClass ClassOf(ESymbolCategory c) {
  switch (c) {
    case ESymbolCategory.Future: case ESymbolCategory.FutureRolover: return EAssetClass.Future;
    case ESymbolCategory.Stock: return EAssetClass.Stock;      // ETF = Stock (rev-3 Q1)
    case ESymbolCategory.Forex: return EAssetClass.SpotFx;
    case ESymbolCategory.Crypto: return EAssetClass.Crypto;
    case ESymbolCategory.Index: return EAssetClass.Index;
    case ESymbolCategory.FutureOption: case ESymbolCategory.StockOption: case ESymbolCategory.CurrOption:
    case ESymbolCategory.IndexOption: case ESymbolCategory.MutualFund: case ESymbolCategory.MoneyMarketFund:
    case ESymbolCategory.Cash: case ESymbolCategory.Bond: case ESymbolCategory.Spread: case ESymbolCategory.CPCSymbol:
    case ESymbolCategory.Composites: case ESymbolCategory.FutureRoot: case ESymbolCategory.FutureOptionRoot:
    case ESymbolCategory.StockOptionRoot: case ESymbolCategory.Commodity: case ESymbolCategory.Warrant:
    case ESymbolCategory.None: return EAssetClass.Unsupported;
    default: throw new InvalidOperationException("M-INSTR: unmapped ESymbolCategory " + ((int)c).ToString(CultureInfo.InvariantCulture));
  }
}
```

All 23 members are listed by name; the `default` throws on a value added in a later MC build (formatting the int is a log, not a parse). It compiles only in PowerLanguage Editor (no test project references PowerLanguage), so it is checked by the PLEditor compile in PR-M2/M3, not by a test. The MC-free core uses only `EAssetClass`.

4.3 Root resolution (PR-M1a, fixes F1; rule rewritten in rev 2.2)

1. **Clean:** trim; remove a leading `@`; remove a `#…` continuous suffix; uppercase invariant; remove every `/` ("EUR/USD" → "EURUSD"). In PR-M1a (`_MtfMath.SymbolRoot`, MC-free, no class) the `/` removal applies to every name; **PR-M2 gates it to Spot FX** (`SpotFx`) once the resolver knows the class.
2. **Root:** take the leading digits plus the run of letters that follows them (stop at the first character that is neither). If the letter run has **≥ 2 letters**, its last letter is a month code (`FGHJKMNQUVXZ`) and the character right after the run is a digit, drop that last letter. Nothing else is removed or kept: exchange suffixes (`:XCME`), spaces, year digits and `.`-classes fall away because the run stops there. Examples (MI-T2): `ESZ2024` → `ES`; `'ES 12-24'` → `ES`; `ESZ24:XCME` → `ES`; `ESZ4` → `ES`; `CLZ2026` → `CL`; `MESZ26` → `MES`; `MESH27` → `MES`; `@6E` → `6E`; `6EZ26` → `6E`; `6B` → `6B`; `EUR/USD` → `EURUSD`; `BRK.B` → `BRK`; `Z4` → `Z` (one-letter run: no month drop). (Rev 2.1 steps 3–4, "name minus a trailing month + 1–2 digit year", left `ESZ2024` unchanged; the code on `6853398` (main before #90) L109–119 already used the letter-run form but drops leading digits and does not remove `/`.)
3. **MC's stored root is primary (PR-M2 resolver; rev 2.3, Ettienne; CONFIRMED by Ettienne `_RootDump` 2026-11-03, relayed by Coder).** MC stores the root itself: `Bars.Info.ASymbolInfo2.SymbolRoot` (PG §4.2.3, `/workspace/research/pg11.txt` L901; struct L3828–3867, field L3833). dxFeed **populates it for futures and stocks** (@ES → `ES`, @CL → `CL`, @6E → `6E`, SPY → `SPY`; §4.4.3). The M2 resolver takes it as the **primary** root (trimmed, uppercase invariant); the #90 parser (`_MtfMath.SymbolRoot`, steps 1–2) is a **pure fallback**, used only when `(s.UnsupportedFields & MTPA_SymbolFields.eMTPA_SF_Root) != 0` (typed; also covers `ALL` = -1; `MTPA_SymbolFields` is not `[Flags]` but its values are powers of two, so the bitwise test is valid) **or** `string.IsNullOrWhiteSpace(s.SymbolRoot)`. Never a string or `ToString()` test on the enum (R1). When both roots are non-empty

and `!_MtfMath.SameRoot(mcRoot, parsedRoot)`, log one **Warning** `INSTR root-crosscheck mc(<mcRoot>) parsed(<parsedRoot>) name(<Info.Name>)`; **MC's root wins** (it feeds presets, roll inheritance and the same-root checks). The INSTR line records `rootSrc=mc|parsed`. PR-M1a (`_MtfMath`, MC-free, #90) keeps the parser only; it has no access to `Info`. **MC-internal cross-check (every stream, before presets):** `s.PriceScale × Info.PriceScale = 1` within ±1e-9 relative (ES/CL/SPY 0.01 × 100; 6E 1E-05 × 99999.999999999985 = 0.99999999999999985) and `s.BigPointValue == Info.BigPointValue`. On a mismatch: Error + throw `mismatch:mcPriceScale` / `mismatch:mcBigPointValue` with a clear message, e.g. `M-INSTR: <ENGINE> cannot run on @6E (class Future, root 6E, sessions Default): mismatch:mcPriceScale (ASymbolInfo2.PriceScale 1E-05 × Info.PriceScale 99999.999999999985 = <x>, expected 1 ±1e-9). Check the symbol's settings in QuoteManager.` (§6.3). `MinMove` is equal in all four rows and is logged, not thrown on.

4. **Class-aware handling (PR-M2 resolver only):** stocks keep their share-class suffix (`BRK.B` is its own symbol; the M1a root `BRK` is used only for the MTF/DVP same-root checks), and the `/` gate of step 1 applies to Spot FX only. Presets (PR-M2) match the step-2 root exactly; the rev 2.1 longest-prefix search is dropped because step 2 already isolates the root (`MES` ≠ `ES`, `6E` ≠ `6B`).
5. Other classes when the fallback applies (MC gives no root or flags it): the cleaned name with the same step 2 (crypto `BTCUSD` → `BTCUSD`, `AAPL` → `AAPL`).
6. An empty root (MC and parser both empty) is an error (`RootError` and roll inheritance never compare empty roots).
7. Presets match on root **and** class ("ES" as a stock never gets the ES futures preset).
8. **Cross-check (futures with a preset):** detected tick = `MinMove/PriceScale` and `BigPointValue` equal the preset within 1e-9 relative, else Error + throw `mismatch:tick` / `mismatch:bigPointValue`. The tolerance covers MC's binary priceScale: @6E reports `priceScale(99999.999999999985)` with `minMove(5)` (`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`), tick 5.0000000000000007e-5 vs 0.00005, relative error ≈ 1.5e-16 (**CONFIRMED** covered). @ES 25/100 bpv 50, @MES bpv 5, @CL 1/100 bpv 1000, @GC 1/10 bpv 100, @6E bpv 125000 all equal the §8.1 presets.
9. **No preset:** roll rule, contract months, expiry and cost are Unknown; an engine that needs one throws (§6.4 S5).

4.4 VERIFY dump `_InstrDump.Strategy.CS` (PR-M0; typed, no reflection)

Ettienne runs it once on @ES (his live chart, Default sessions), @ES on the "CME Globex" template (the D9a/canary chart), @MES, @CL, @GC, @ZN, @6E, one FX pair, one crypto symbol and one stock, and once in Portfolio Trader. Output: see **§4.4.1** (same directory as the standard MCSignal log, N6 DECIDED). Doubles via `ToString("R", CultureInfo.InvariantCulture)`. Restart MultiCharts after changing any symbol's sessions in QuoteManager (cached until restart, §4.5 setup note).

4.4.1 Output file (N6 DECIDED)

- **Directory = the directory of the standard log file the host already creates.** The resolver on `7dbdb1b` is in `_CustomSignal.Strategy.CS`, nested `LogFileSingleton` constructor **L71–91**: `Path.Combine(Environment.GetFolderPath(SpecialFolder.LocalApplicationData), "MCSignalLogs")` (L79), `Directory.CreateDirectory` (L80), a zero-length write to prove access (L82), and on any exception the fallback `Path.GetTempPath()` (L85–88). The resolved file is exposed as `LogFileSingleton.FilePath` (L107) and obtained via `LogFileSingleton.GetInstance` (L93; used at L371 and L395). The same two-step rule is duplicated inline in `TradeJournal` (constructor L168–184, dir L176, fallback L182, `MCJournal_`/`MCEvents_` files).
- **Reuse, no new path.** `_InstrDump` derives from `_CustomSignal` (precedent: `_MtfProbe : _CustomSignal`, `_MtfProbe.Strategy.CS` L89). PR-M0 adds one accessor to `_CustomSignal`, no behaviour change: `protected string LogDirectory() { if (m_logger == null) m_logger = LogFileSingleton.GetInstance(null); return Path.GetDirectoryName(m_logger.FilePath); }` (same lazy pattern as `traceAlways`, L395). It therefore returns the directory of the log file actually created, including the `%TEMP%` fallback. No `const` path, no `C:\Share` path, no new environment lookup in `_InstrDump`.
- **Name**, following the existing `MCSignal<yyyyMMdd-HHmmss>-<pid>.csv` (L76) / `MCJournal_<runId>.csv` / `MCEvents_<runId>.csv` pattern: `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt`, timestamp `DateTime.Now` with `CultureInfo.InvariantCulture`, `<symbol>` = `Bars.Info.Name` with every `Path.GetInvalidFileNameChars()` character and `/` replaced by `_` (`@ES` stays `@ES`; `EUR/USD` → `EUR_USD`). One file per symbol per run (Portfolio Trader: one per symbol).
- **Writing:** one `File.AppendAllText(path, line + Environment.NewLine, new UTF8Encoding(false))` per line (UTF-8 without BOM, as the log, L66), doubles `ToString("R", CultureInfo.InvariantCulture)`. On the first line `_InstrDump` writes one trace line to the standard log, `INSTRDUMP file(<full path>)` (same form as `JOURNAL file(...)`, L2407), and one `Output.WriteLine` with the path. IO errors go to Output and never stop the strategy.

- **Collecting (PowerShell one-liners, no variables).** List the dumps in both possible directories, newest last: `Get-ChildItem -LiteralPath (Join-Path ([Environment]::GetFolderPath('LocalApplicationData')) 'MCSignalLogs'), ([System.IO.Path]::GetTempPath()) -Filter 'MCInstrDump_*.txt' -ErrorAction SilentlyContinue | Sort-Object -Property LastWriteTime | Format-Table -Property LastWriteTime, Length, FullName -AutoSize` Make a Desktop folder for them: `New-Item -ItemType Directory -Force -Path (Join-Path ([Environment]::GetFolderPath('Desktop')) 'InstrDump')` Copy every dump into it: `Get-ChildItem -LiteralPath (Join-Path ([Environment]::GetFolderPath('LocalApplicationData')) 'MCSignalLogs'), ([System.IO.Path]::GetTempPath()) -Filter 'MCInstrDump_*.txt' -ErrorAction SilentlyContinue | Copy-Item -Destination (Join-Path ([Environment]::GetFolderPath('Desktop')) 'InstrDump') -Force` Then send the `InstrDump` folder (or attach the files to the PR-M0 thread); they are committed as `tests/InstrumentTests/fixtures/MCInstrDump_*.txt` (data only; nothing in M1 depends on them).

- **A1 (compiles today):** `name, cat (log), catInt, exch, tz (log), minMove, priceScale, bpv, bvo (log), res.Type (log), sessionName`. Description, PointValue and Resolution.Size are already used in compiled code (`_EconomicCalendar` L231/L243/L679, `_CustomSignal` L966/L2245) and are not dumped (`Info.SessionName`, confirmed-PG; record whether it prints "Default" or the template name).

- **A2 Sessions at StartCalc vs first bar:** print `Bars.Sessions` (`n`, then each `StartDay StartTime -> EndDay EndTime`) **in StartCalc and again on the first CalcBar**, and the first bar's `Bars.Time[0]` and `Bars.LastBarInSession`. Records: populated at StartCalc (yes/no/null), identical at the first bar, multiple sessions per day kept or collapsed (forum: fixed in MC .NET 9.0). **Result: CONFIRMED** populated and identical in all 13 runs; multiple sessions per day kept (GLOBEX: 9 entries, 17:00–15:15 and 15:30–16:30 each weekday).
- **A3 Session and bar zone:** run @ES and @CL each twice, Format Instrument → Time Zone = **Exchange**, then **Local**. Compare the same bar's `Time[0]` and the printed session times between runs: (i) whether `SessionObject` times follow the chart zone (staff: "main data series" zone); (ii) the exchange offset for NYMEX/COMEX (settles row 4, L2621). **Result: CONFIRMED** (i): session and bar times follow the chart zone (@ES 17–16 Exchange, 18–17 Local); (ii) @CL and @GC show the same 18:00–17:00 either way, see row 4.
- **A4 Multi-series / Portfolio:** with Data2 on a different template, print Data2's sessions (`BarsOfData(2).Sessions`) to confirm per-series sessions; in Portfolio Trader, print each symbol's sessions and `Info.TimeZone` (PT time zone is one global setting). **Result:** per-series sessions **CONFIRMED** (Data1 Default n=5, Data2 GLOBEX n=9, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`); Portfolio Trader **OPEN** (only @ES written, ds2–10 absent, `01M6HSDBFH56AB94QP1SZ4NFP7.txt`).
- **B (resolved by `_RootDump`, rev 2.3; Ettienne `_RootDump` 2026-11-03, relayed by Coder):**

- **B1 expiry:** `ASymbolInfo2.SymbolExpiry` is a **placeholder** on continuous futures (2038-01-01 on @ES, @CL, @6E) and stocks (1900-01-01 on SPY). It is used **only when strictly after 1900-01-01 and strictly before 2038-01-01** (i.e. a dated contract); otherwise expiry and roll come from the roll calendar (§8.1) and the roll detector. `Info.Expiration` stays VERIFY and is not needed.
- **B2 currency:** `ASymbolInfo2.SymbolCurrency` (typed `MTPA_MCSymbolCurrency`): **CONFIRMED** USD for all four; non-USD throws (Q8). `StrategyCurrencyCode` is not needed.
- **B3 cross-checks:** `SymbolRoot`, `SymbolExchange`, `SymbolCategory`, `MinMove`, `PriceScale`, `BigPointValue` **CONFIRMED** (§4.4.3); `PriceScale` convention: struct = 1/`Info.PriceScale` (the PG "for ES = 1/100" example, L3853, is the struct's); the resolver check is §4.3 step 3. `SymbolCategory` is typed (`MTPA_MCSymbolCategories`) and is logged, not used for classification (§4.2 stays on `Info.Category`).
- `_RootDump` was Coder's throwaway; not committed. Its four rows are the MI-T2/MI-T3 fixtures (plain strings/ints/doubles in the MC-free core).

`{i.Category}` etc. are formatted only for the human log; nothing parses them (R1).

4.4.2 PR-M0 dump results (Ettienne's runs, 2026-11-03 17:28–17:43 ET)

Thirteen fixture files from the PR-M0 `_InstrDump` build (rev 2.3: the 6E rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` replaces the discarded `01M6HSDCCJZXJHNB165KSF6Z5B.txt`; the latter is still listed in `index.txt` but is not committed), indexed in `/workspace/instrdump/index.txt` (to be committed as `tests/InstrumentTests/fixtures/`; the file names are the upload names, not the `MCInstrDump_*` names). Not run yet: @ZN, an FX pair, a crypto symbol (B1/B2 resolved by `_RootDump`, §4.4.3).

File	Run	Facts read from it
`01M6HSDCPGEWFS9VAYZG2C808V.txt`	@ES, Exchange, Default	n=5, Sun 17:00 → Mon 16:00 … Thu 17:00 → Fri 16:00; same at StartCalc and first bar; session-end bars 16:00
`01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`	@ES, Exchange, GLOBEX template	`sessionName(CME: Equity Index Futures [E-mini] (GLOBEX))`; n=9 (17:00–15:15 + 15:30–16:30); **session-end bars only at 16:00, none at 15:15** (R14)
`01M6HSDC526NQSHVTNJG9AQ7A7.txt`	@MES, Exchange, Default	as @ES, bpv 5
`01M6HSDDB20HAY23SAABKQ0MQ5.txt` / `01M6HSDCMNQG1HPR0B9XQWC4S2.txt`	@CL Exchange / Local	18:00–17:00 in both; bars end 17:00 in both; minMove 1, priceScale 100, bpv 1000
`01M6HSDC20TJ572NEKZBKKRBWD.txt`	@GC, Exchange	18:00–17:00, bars end 17:00; minMove 1, priceScale 10, bpv 100
`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`	@6E, Exchange (rerun 2026-11-03 18:05 ET, after switching @6E to the Exchange & ECN sessions and restarting MC)	`cat(Future)`, exch CME, priceScale 99999.999999999985, minMove 5, bpv 125000; Default session Sun–Thu 17:00 → next day 16:00 (Exchange), first bar 17:06, session-end bars at 16:00 (`lastBarInSession(True)`): **matches ES**. Chart has `bvo(TickCount)`: a futures chart on Tick Count must throw under R7 (volume basis), so this fixture is for **instrument and session facts only**, never a volume-engine fixture. Replaces `01M6HSDCCJZXJHNB165KSF6Z5B.txt` (misconfigured custom sessions, **discarded**; not a fixture)
`01M6HSDBFJ9WRF71G51B82VKCH.txt`	@ES, Local	18:00–17:00; first bar 18:07, session ends 17:00 (= Exchange run + 1 h)
`01M6HSDBG7NET9EVPFJ8Z43MR5.txt`, `01M6HSDBWZ22Q03T3AV5GWP4CY.txt`, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`	@ES Data1 Default + @ES Data2 GLOBEX (three identical runs apart from run id)	Data1 n=5, Data2 n=9 (per-series sessions); Data1 16:00 bar of 2026-10-26 logged as a session end twice (R14 idempotence)
`01M6HSDBFH56AB94QP1SZ4NFP7.txt`	Portfolio Trader, @ES + @CL, Local	`maxDataStream(10)`, ds2–10 `absent`; only the @ES dump was written (OPEN)
`01M6HSDC62GCMK5ZANKB61PD5T.txt`	SPY, Exchange	`cat(Stock) catInt(2)`, exch ARCA, `bvo(TickCount)`, Mon–Fri 04:15–20:00, session ends 20:00

CONFIRMED: (1) `Bars.Sessions` is filled at StartCalc and identical on the first bar (all 13); (2) session and bar times follow the chart time zone (@ES 17–16 Exchange vs 18–17 Local; @CL and @GC 18–17 either way); (3) sessions are per data series and multiple sessions per day are kept; (4) @6E priceScale is within the 1e-9 tolerance; (5) SPY is Stock / TickCount / 04:15–20:00; (6) `SessionName` returns the chart's template name. **NEW MC behaviour:** `LastBarInSession` did not fire at the GLOBEX 15:15 break, only at 16:00 (and on the 15:30–16:30 session it fired at 16:00, before the listed 16:30 end, because data stops at 16:00) → R14, MI-T16. **OPEN (awaiting Ettienne):** (a) Portfolio Trader with @ES + @CL wrote only the @ES dump (`maxDataStream(10)`, ds2–10 absent): PT per-symbol sessions and `Info.TimeZone` for the second symbol remain unverified (does `_InstrDump` run per symbol in PT, or did @CL not start?). Ettienne's rerun of @6E is also pending (below). **CLOSED (rev 2.3): @6E zone.** Ettienne's QuoteManager had @6E on **Use Custom Sessions**, Sun–Thu 18:00 → next day 17:00, with Session TimeZone = **Exchange (CT)**: ET hours entered as CT. A user configuration error, not MC behaviour (CME 6E trades 17:00–16:00 CT like ES, research `/workspace/m-instr/mc-session-templates-research.md` "Observation: 6E", https://www.cmegroup.com/trading/why-futures/welcome-to-cme-fx-futures.html). He switched @6E to the Exchange & ECN sessions and restarted MC; the rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` reads Sun–Thu 17:00 → 16:00 Exchange, first bar 17:06, session ends 16:00, matching ES. It is the 6E fixture; `01M6HSDCCJZXJHNB165KSF6Z5B.txt` was misconfigured and is discarded (dropped from the fixture list).

4.4.3 `_RootDump` results (Ettienne `_RootDump` 2026-11-03, relayed by Coder; dxFeed)

Symbol	root	exch	cat	ccy	SymbolExpiry	minMove	PriceScale struct vs Info	BPV struct vs Info	UnsupportedFields
@ES	ES	CME	FUTURE	USD	2038-01-01	25	0.01 vs 100	50 vs 50	0
@CL	CL	NYMEX	FUTURE	USD	2038-01-01	1	0.01 vs 100	1000 vs 1000	0
@6E	6E	CME	FUTURE	USD	2038-01-01	5	1E-05 vs 99999.999999999985	125000 vs 125000	0
SPY	SPY	ARCA	eMTPA_MCSC_STOCK	USD	1900-01-01	1	0.01 vs 100	1 vs 1	0

`SymbolName` = `Info.Name` in every case. `MTPA_SymbolFields` values as in §4.1.

4.5 Sessions from the chart (R12; DECIDED Q1/Q2)

What MC does (research doc `/workspace/m-instr/mc-sessions-research.md`). Each data series has its own session setting (Format Instrument → Settings → Sessions = **Default**, i.e. the symbol's QuoteManager sessions, or a named template); in Portfolio Trader each symbol has its own session but the time zone is one global setting. Scripts cannot set sessions. Data outside sessions is filtered out; time bars always break on session boundaries and align to the session start; `Time` is the bar close. Holidays are removed by QuoteManager holiday lists and **cannot be read from code**. **Templates (research `/workspace/m-instr/mc-session-templates-research.md` §1, §3, §7):** a chart uses Default (the symbol's QuoteManager sessions: exchange sessions, custom sessions, a custom template or the symbol dictionary) or a named template; templates, built-in ones included, are editable in **QuoteManager → Tools → Session Templates** (Add/Edit/Delete, rows with Session Time and Session End; https://www.multicharts.com/trading-software/index.php?title=Setting_Custom_Session_Templates, https://www.multicharts.com/discussion/viewtopic.php?t=5890), but not from a script (staff, https://www.multicharts.com/discussion/viewtopic.php?t=53293). **"Combined" templates mean two sessions per day** (overnight + day, two daily bars; https://www.multicharts.com/discussion/viewtopic.php?t=45183), not one merged 23-hour session; "Open Outcry"/"Open Auction" templates describe pit sessions closed in 2015 and survive only as RTH-style windows.

Setup note (MC caches sessions). MultiCharts caches QuoteManager symbol sessions until a restart: after changing a symbol's sessions in QuoteManager, **restart MultiCharts** before running charts, dumps or tests (the @6E rerun, §4.4.2, needed it).

Environment caveat (not an engine rule). The built-in **"CME: Equity Index Futures [E-mini] (GLOBEX)"** template is stale: its 15:15–15:30 CT gap is the halt CME removed on 27/28 June 2021 (https://www.cmegroup.com/notices/electronic-trading/2021/06/20210621.html), and its 15:30–16:30 row matches the hours **before November 2012** (2012–2021 sessions ended 16:15, https://www.cmegroup.com/tools-information/lookups/advisories/market-data/20121112.html; pre-2012 sheet https://www.dormantrading.com/TraderTools/CMEeminisp500.pdf); current ES hours are 17:00–16:00 CT (https://www.cmegroup.com/markets/equities/sp/e-mini-sandp500.html). This template is on the **D9a/canary chart** (§4.4), so that chart has a session boundary at 15:15 and a listed end at 16:30. Engines **take sessions as given** (R12) and do not correct or warn about it; changing the canary chart's template is Ettienne's environment decision and would change MI-T15 baselines.

Reading. `SessionReader` (host) reads `BarsOfData(n).Sessions` into the MC-free struct `_MtfMath.Session` (#88 head, L47–52: `StartDay, StartTime, EndDay, EndTime`) **in StartCalc** (populated there, CONFIRMED §4.4.2) and **re-checks on the first CalcBar of each stream** (must be identical, else Error + throw `sessions:changed`, S1). Session ends follow R14 (from the list's `EndTime`s, not `LastBarInSession` alone). Null or empty sessions → throw `sessions:none` for every session-using engine (§6.4 S1). Template name: `Info.SessionName` on the INSTR line, log only. Whether the chart uses Default is not exposed (not documented) and is not checked: the chart's template is the truth.

Zone. Session times are in the bar time zone of the stream, i.e. the chart's Format Instrument time zone (**CONFIRMED** A3, §4.4.2: @ES 17:00–16:00 on Exchange, 18:00–17:00 on Local, bar times shift by the same hour; @CL/@GC identical either way). No conversion is needed before deriving. ET-anchored logic that remains (EVENT rows, the frozen G0/REGGATE trial windows) converts bar time to ET as today: Local → `TimeZoneInfo.Local`, GMT → UTC, Exchange → the `exchangeTimeZoneId` map (L2614–2626, kept only for this; its NYMEX/COMEX entry stays Eastern: A3 shows MC's exchange time for @CL/@GC equal to ET on Ettienne's setup, row 4).

Derived fields (MC-free, `<instr-core>`): - `tradingDayKey(t)` = the calendar date, in session time, of the **end** of the session that contains bar close `t` (start exclusive, end inclusive). Several sessions ending on the same date share the key (e.g. the GLOBEX template 17:00–15:15 + 15:30–16:30, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`). On the CME Globex template (17:00–16:00 CT) this equals today's `GlobexTradingDayEt` (18:00 ET roll) for every bar; on a 24×5 FX template ending 17:00 ET it equals today's ET+7h rule; on a 00:00 UTC crypto template it equals the UTC date. A bar outside every session (should not occur, MC filters) → Error + throw `sessions:bar-outside`. - `dayOpen` = start of the first session of the trading day (TOD minute-of-session; replaces 18:00 ET at VPFUT L489–493 and the default at L949). - `sessionEnd` = end of the last session of the trading day, from the list (R14; on the GLOBEX template 16:30 even though the last bar closes at 16:00) (FLOW tick-chunk boundary, replaces `TickChunkBoundaryEt` 17:00 and the rev-1 `rollTime − 1h`). - `hasOvernight` = some session of the trading day crosses midnight in session time (`StartDay != EndDay`). Globex 17:00 CT Sun → 16:00 CT Mon: true; an 08:30–15:15 day session: false. For 24-hour single-day templates (e.g. 00:00–24:00 crypto) the value is fixed in PR-M4 from the A2 output (**VERIFY**). - `rthSplit` = the RTH segment inside the trading day: **when the chart's sessions are day-only (`!hasOvernight`), the chart session is the RTH**. For a 24-hour chart (N1 DECIDED): ES/MES/NQ/MNQ use today's equity-index split from `_CmeSessionClock` (RTH 09:30, value end 16:00, session flag 16:15 ET), logged `rth src=EquityIndexLegacy`; any other root on a 24-hour chart has `rthSplit = null` (S7). No RTH window in any preset. - `sessionsPerWeek`/`sessionsPerYear`: annualisation stays a class constant (252 futures/stocks, 365 crypto, 260 FX **VERIFY**), not derived.

Multi-stream. Each stream has its own derived fields. The #86/#88 DV/MTF Warning when Data2..N sessions differ from Data1 (`checkHtfSessions`, #88 head L1802–1822) stays unchanged, including its MC-1439 / Realtime-History Matching caveat: with RHM on and different session ends, the strategy stalls after the shorter session's end until the auxiliary series prints. In Portfolio Trader, mismatched bar-close times across symbols can make the portfolio money-management signal run several times per bar (MC staff); M-PORT behaviour is unchanged, and the INSTR line shows each symbol's sessions so a mismatch is visible.

Removed because they existed only for hardcoded times (PR-M4 unless noted): `_CustomSignal` L1888–1889 (`"Eastern Standard Time"` + `GlobexTradingDayEt` for the roll key → `tradingDayKey`); `autoTradingDay` L2581–2587 + L2462 (→ `tradingDayKey`); VPFUT `GlobexDate` key L361 and TOD 18:00 L489–493; `_CmeSessionClock` Globex open/close/halt defaults L48–61 as trading-day inputs; `TickChunkBoundaryEt` (PR-M6b). The exchange-zone map stays only for ET conversions (above). Holiday calendars stay (they are not session times; §6.4 S4, PR-M8).

ES bit-identity. On the CME Globex template (D9a and canary charts, `_DvpEngineTest` L28–48, `_TestGate0` L42) every derived day key, day open and session end equals today's constant. On a chart with another template (e.g. Ettienne's live @ES Default sessions) results follow that chart; this is the intended behaviour of the decision and is logged on INSTR, not warned.

5. Data model

5.1 `InstrumentProfile` (immutable; MC-free core `<instr-core>` in `_InstrumentProfile.Function.CS`)

Field	Type	Source	Notes
`assetClass`	`EAssetClass { Future, Stock, SpotFx, Crypto, Index, Unsupported }`	detect (§4.2)	
`root`	string	`ASymbolInfo2.SymbolRoot` (CONFIRMED), parser only as fallback (§4.3 step 3)	"ES", "6E", "EURUSD", "BTCUSD", "AAPL"; never empty
`presetId`	string or null	preset	
`tick`	double	`MinMove/PriceScale`	> 0 else throw
`bigPointValue`	double	`Info.BigPointValue`	currency per 1.0 price move (renamed from rev-1 `pointValue`; MC's `Info.PointValue` is another member)
`tickValue`	double	tick × bigPointValue	
`currency`	ISO code	`ASymbolInfo2.SymbolCurrency` (CONFIRMED, §4.4 B2)	non-USD throws (Q8)
`sessions`	`_MtfMath.Session[]`	`Bars.Sessions` (§4.5)	read on the first bar
`sessionName`	string	`Info.SessionName`	log only
`barZone`	`RequestTimeZone` + resolved tz id	`Info.TimeZone` (+ map for Exchange)	ET conversions only
`tradingDay`	derived from `sessions`	§4.5	key = session-end date
`dayOpen`, `sessionEnd`, `hasOvernight`	derived	§4.5	
`rthSplit`	(start, end) or null, + `rthSource ∈ {ChartDaySession, EquityIndexLegacy, None}`	§4.5, N1	
`calendar`	`ECalendar { CmeEquityIndex, Nyse, Observed, None }` (+ CME product groups when verified, PR-M8)	preset	holidays/early closes only; no session times
`contractMonths`	month mask	preset	
`expiry`	`EExpiryRule` + params	preset; a dated contract's `SymbolExpiry` when strictly inside (1900-01-01, 2038-01-01) (§4.4 B1)	§8.2
`roll`	`ERollRule` + params	preset	§8.2
`sessionsPerYear`	int	class	252 / 365 / 260 (FX **VERIFY**)
`cost`	`CostModel { roundTripCents (long), source }`	`InstrumentConfig.costs`	Unknown when absent
`costFloorTicks`	int	§9 (integer cents)	
`cluster`	string	preset	same strings as L1611–1614
`volumeBasis`, `volumeQuality`		`Info.BVO` + class	R7
`dataNotes`	list of (from, what)	`InstrumentConfig.dataNotes`	Q7
`sizingUnit`	Contracts / Shares / Units	class	

5.2 Presets: exchange facts only (no times)

`InstrumentPresets`: static table in `<instr-core>`, one row per root (§8): tick, bigPointValue, currency, contractMonths, expiry, roll, calendar, cluster, source. Adding a root = a one-row PR with sources.

5.3 `InstrumentConfig` (protected on `_CustomSignal`; no inputs)

- `enabled` (log-only until PR-M4; then applied).
- `overrides`: typed `InstrumentPresetRow`, null by default (fix an exchange fact without a code PR; contains no session fields).
- `costs`: root → round trip in **cents** (`long`), default `{ES: 1700, MES: 250}`.
- `dataNotes`: root → notes.
- `Validate()`, `Canon()` as in existing configs; `profileHash` outside every `configHash` (rev-3 Q7).

6. Logging, application and refusal

6.1 Lines (`engineSink(level)`, one `File.AppendAllText` per line, InvariantCulture)

- `INSTR ds(1) name(@ES) cat(Future) class(Future) root(ES) preset(ES) exch(CME) tz(Exchange→Central Standard Time) sessionName(<Info.SessionName>) sessions(n=1 Sun 17:00→Mon 16:00; …) dayKey(SessionEnd) dayOpen(17:00) sessionEnd(16:00) overnight(yes) rth(<start>-<end> src=EquityIndexLegacy|ChartDaySession|None) tick(0.25) bpv(50) tickValue(12.5) ccy(USD) bvo(Volume) calendar(CmeEquityIndex) months(HMUZ) expiry(ThirdFridaySoq) roll(EquityIndexCustomary) cost(17.00 src=Ettienne 2026-10-31) floor(14) cluster(EQIDX) profileHash(…)` at **Info**, once per stream after the first-bar session read.
- `CFG <ENGINE> applied{…} undefined{output=reason;…} degraded{mode}` at Info, once per engine.
- `SKIP <ENGINE> reason(<code>) …` at **Info**.
- Mismatches go out at **Error** before the throw.

6.2 Application

`buildProfiles()` runs in `StartCalc` for instrument fields; session-derived fields are filled on the first CalcBar of each stream, and engines whose configuration needs them (TOD, AVWAP session date, FLOW chunk plan, RANGE/WEEKLY session sources, VPFUT day key) finish configuring there, before their first update. Engines get the profile through their existing config objects; engines never read `Bars.Info` or `Bars.Sessions` directly.

6.3 Exception format

`M-INSTR: <ENGINE> cannot run on <name> (class <c>, root <r>, sessions <sessionName>): <reason>. <action>.` Reason codes: `class`, `unknown-field:<f>`, `tick`, `mismatch:<f>` (incl. `mcPriceScale`, `mcBigPointValue`), `volume-basis`, `currency`, `sessions:none`, `sessions:changed`, `sessions:bar-outside`, `sessions:no-rth-split`, `sessions:window-not-covered`, `calendar`.

6.4 R13 applied: every throw, degraded mode and undefined output

Throw rules | # | Condition | Engines | Action text | |---|---|---|---| | S1 | `Bars.Sessions` null/empty at StartCalc or on the first bar; or StartCalc and first-bar reads differ; or a bar outside every session | every session-using engine (TOD, AVWAP, VPFUT, FLOW, RANGE, WEEKLY, DAILY, DVP, G0, REGGATE, ROLL keying) | "check the chart's session template" | | S2 | An engine whose purpose needs an overnight segment, on a chart with `!hasOvernight` | none on main (every overnight-specific item is an output, see U1–U2) | — | | S3 | A frozen pre-registered window (G0 09:30–16:00 ET, `_Gate0` L474/L614/L1037…; REGGATE 09:30–16:00 ET, `_RegimeGate` L45–48) is not inside the chart's sessions (converted to ET) | G0, REGGATE | "use a session template that covers 09:30–16:00 ET" | | S4 | An engine needs a holiday or early close **in advance** and the root has no verified calendar | REGGATE (early-close exit), G0 (early flat), VPFUT `HaltEt/RthCloseEt/IsGlobexClosedDay` hooks (L1635–1671), `_CmeSessionClock` holiday use | "add a verified calendar for (PR-M8)" | | S5 | A required preset field is Unknown (`roll`, `contractMonths`, `expiry`, `cost`) | ROLL Table/SeriesWindow, EVENT expiry kinds, DVP roll table, COST/cost floor | "add a preset row / cost for " | | S6 | Class Unsupported in the rev-3 matrix | per matrix | rev-3 reason text | | S7 | An engine needs `rthSplit` and it is null (24-hour chart, root not ES/MES/NQ/MNQ; N1) | VPFUT RTH profile/value window/IB, AVWAP `rth`, WEEKLY `RthSessions`, DVP RTH features | "use a day-session template for " | | S8 | tick ≤ 0; detected ≠ preset; futures on Tick Count (R7); non-USD (Q8) | all | — |

Approved degraded modes (only these; each from a DECIDED answer): D1 VP on FX/CFD/crypto as visual/level only, nothing feeds Holm (rev-3 Q5); D2 G0 on Index price-only parts (rev-3 matrix note 6; G0 is ES-canary only in practice); D3 EVENT on USD FX pairs, US events as a filter (rev-3 Q11); D4 FLOW on crypto, single venue, Warning (rev-3 reason 12); D5 PriceAction on Bond/MF/Unknown (rev-3 reason 2); D6 FLOW bid/ask features undefined before a root's `dataNotes` bid/ask start date (Q7, accepted 2026-11-02); D7 TOD on a root without a verified calendar: no current-day half-day flag, observed short days excluded from baselines, one Info line (N4 DECIDED). Rev 1's "Degrade" on RTH-only charts, on unknown early closes and on unconfirmed RTH windows is **withdrawn**. A new degraded mode needs Ettienne's approval.

Undefined outputs (NaN / false, one Info line, not a degrade) | # | Output | When | |---|---|---| | U1 | VPFUT overnight stats `ethHigh/ethLow/ethVolume` (L378) | `!hasOvernight` (N3 DECIDED) | | U2 | VPFUT `IsAsia()/IsEurope()` helpers (L244–245) | `!hasOvernight` (N3 DECIDED) | | U3 | DVP `rthShare` | `rthSplit == null` (already NaN on the generic base) |

Post-hoc calendar use without a verified table (correct, not degraded): `tradingDayCount` (L3228–3240) counts observed trading days (days that have bars) instead of NYSE days for roots without a table; `calendar = Observed`.

7. Per-engine migration

Engine	Change	ES result (CME Globex template)	PR
CS host	L2462 typed; `autoTradingDay` and L1888–1889 → `tradingDayKey`; tick fallback → throw; `_NyseHolidays` for stocks, `p.calendar` otherwise; exchange map kept for ET conversions	identical	M1b, M4
CLOCK / VPFUT	Day key, day open, Globex open/close come from session data; RTH split per N1 (ES: today's constants, `rthSource=EquityIndexLegacy`); value window (09:30, 16:00] unchanged for ES; Asia/EU/NY windows per N5; holiday hooks per S4	identical	M4
CAL	ES keeps `_CmeHolidays` (both `UseCmeHolidayTable` flags, VPFUT L1627 and clock L81); NYSE for stocks; other roots `Observed` for post-hoc use and S4 throws for in-advance use until a verified table lands	identical	M8
ROLL	`_RollDetector` takes months/expiry/roll from the profile; F2; `gapTicks` stays 8 (Q4)	identical	M5

Engine	Change	ES result (CME Globex template)	PR
EVENT	`RollDate/ExpiryDate/SessionsToExpiry` from `p.expiry/p.roll/p.contractMonths`; event classes per asset class	identical	M5
G0	Remove `tick = 0.25` (L357; throw if 0) **and update all 35 `new Gate0Engine` constructions in `tests/Gate0Tests/Program.cs` to set `tick = 0.25`** in the same PR (none sets it today); `"ES"` (L1204) → `p.root`; S3 coverage check; windows frozen	canary hash identical	M6a
FLOW	Tick-chunk boundary = `p.sessionEnd` (CME 16:00 CT = 17:00 ET, today's value); spot FX stays Unsupported (rev-3 reason 11); FX futures get their own session end	identical	M6b
DAILY / REGIME	Annualisation reads `p.sessionsPerYear` (252)	identical	M6c
LEVEL / AVWAP	Remove the 0.25 defaults (`_LevelStudy` L79, `_AvwapManager` L153); both hosts set them (L1410, L1976); AvwapTests already sets 0.25 (L17); `_AvwapManager` L195 already rejects ≤ 0	identical	M6d
RANGE / WEEKLY / PSTATS	`ringBars` **literals kept** (`_RangeProfile` L31 2880, `_ProfileStats` L47 1440; bar counts, not minutes; no derivation); WEEKLY `RthSessions` needs `rthSplit` (S7), else the user picks GlobexSessions; `_ProfileStats` 4-tick note in CFG	identical	M6e
PORT	`PortfolioCluster` returns `p.cluster`, same strings	identical	M6f
DVP	root from `p.root` (F1 done in M1a); roll table from the profile replaces `root == "ES"` (L213/230) and host loops (L119–121)	identical (post-#88 baseline)	M6g
REGGATE / `_Test6`	ES only (N2); windows frozen + S3; `tradingDay` hash text from an explicit `switch (ETradingDayKey) { case Globex18ET: return "Globex18ET"; … }` and `RollDetection` via an explicit switch with the same spellings (L62, `_Test6` L105–106); roll table from the profile (`_Test6` L100–102)	hash identical	M6h
SIZE / COST	`unitNotional` = entry × `p.bigPointValue`; cost floor = **new opt-in behaviour** (§9)	identical (off by default)	M7
MTF	`SymbolRoot` resolves via §4.3; doc fix (L28)	identical for ES	M1a
LEGACY	F3a/F3b only; others a one-line note	—	M1c

8. Presets: exchange facts only, no session times (sources read 2026-11-02)

8.1 Futures

Root	Tick	Big point value ($/1.0)	Tick $	Months	Expiry rule	Roll rule (Q5 DECIDED method)	Calendar	Cluster	Source
ES	0.25	50	12.50	HMUZ	3rd Fri 09:30 ET SOQ, back to the NYSE business day	Mon before the 3rd Fri (3rd Fri ≥ 2022-04-01), else Thu 8 days before (`_EventCalendar` L299–303)	CmeEquityIndex (`_CmeHolidays`)	EQIDX	CME ES specs; code
MES	0.25	5	1.25	HMUZ	as ES	as ES	CmeEquityIndex	EQIDX	CME Micro E-mini FAQ
NQ	0.25	20	5.00	HMUZ	as ES	as ES (**VERIFY** against the vendor splice)	CmeEquityIndex	EQIDX	CME Micro FAQ
MNQ	0.25	2	0.50	HMUZ	as ES	as NQ	CmeEquityIndex	EQIDX	CME Micro FAQ
CL	0.01	1000	10.00	all 12	3 business days before the 25th of the prior month (4 if the 25th is not a business day)	2 business days before LTD (**VERIFY** vs vendor splice)	Observed until verified (S4)	ENERGY	CME CL specs
GC	0.10	100	10.00	active G J M Q V Z	third-last business day of the contract month	business day before first notice (**VERIFY**)	Observed	METALS	CME GC specs
ZN	0.015625	1000	15.625	HMUZ	7 business days before the last business day of the month	business day before first notice (**VERIFY**)	Observed	RATES	CME ZN specs
6E	0.00005	125,000	6.25	HMUZ (serials listed)	2nd business day before the 3rd Wed	Mon of the expiry week (**VERIFY**)	Observed	FX_EUR	CME 6E specs

Notes. No session, RTH, halt or Globex-time column: those come from the chart (R12). Rev 1's RTH reference windows (CL 09:00–14:30, GC 08:20–13:30, ZN/6E 08:20–15:00 ET) and the ES 15:15–15:30 CT halt question are **dropped** (Q2 moot). `BigPointValue` for 6E is **CONFIRMED** 125000 (`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`); ZN is still **VERIFY** (not in the 13 runs); the cross-check (§4.3 step 8) refuses a mismatch. "Observed" calendars serve post-hoc use only; in-advance use throws (S4) until PR-M8 adds a sourced table.

8.2 Rule kinds (typed enums)

`EExpiryRule { ThirdFridaySoq, CL25thMinus3Bd, ThirdLastBd, LastBdMinus7Bd, ThirdWedMinus2Bd }`, `ERollRule { EquityIndexCustomary, BusinessDaysBeforeExpiry(n), BeforeFirstNotice(n), None }`. ES/MES/NQ/MNQ use `ThirdFridaySoq` + `EquityIndexCustomary` = today's `EventRules.RuleExpiryDate/RollDate`.

8.3 Non-futures

Class	Trading day	Calendar	Cost	Notes
Spot FX (12)	from the chart's sessions (a 17:00 ET-ending template gives today's ET+7h day)	Observed	Unknown (not costed until traded)	TickCount volume allowed (R7); DVP, G0, FLOW Unsupported (rev 3)
Crypto (20)	from the chart's sessions	Observed	Unknown	`sessionsPerYear 365`
Stock (2)	from the chart's sessions	Nyse (`_NyseHolidays`)	Unknown	FLOW and VPFUT Unsupported (rev 3). SPY (**CONFIRMED** `01M6HSDC62GCMK5ZANKB61PD5T.txt`): `Stock`, ARCA, Build Volume On = TickCount (tagged `TickCount`, R7, no throw), Default session Mon–Fri 04:15–20:00 Exchange time. That template is day-only, so by N1 its `rthSplit` is the whole 04:15–20:00 chart session, extended hours included; an RTH-only study needs a 09:30–16:00 template on the chart.

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9. Sizing and costs (R8, R9): as the code is on `7dbdb1b`

1. q = 1% risk on the structural stop (CS-4 `sizeForRisk`, L2921), then `maxQty` (L2943) and the notional cap (L2945–2949) **reduce** q (Info when they bind). q never grows.
2. `willArm = m_portOn && bars == Bars` (L2957). - **Not willArm** (M-PORT off, or Data2..N): if q × stop × bigPointValue > 1% × equity the entry is **skipped**, Info "trade skipped" (L2958–2968). - **willArm** (portfolio Data1): the order goes to `portfolioArm` (L2970) and `_PortMM` applies deny-not-reduce. M-INSTR changes nothing here.
3. **Cost floor = new opt-in behaviour (PR-M7)**: `InstrumentConfig.applyCostFloor = false` by default (no input). When on, the stop is widened to `max(structural, costFloorTicks × tick)` before step 2, so the existing skip/deny paths act on the widened stop. Rev-1's claim that main already does this is withdrawn (`costFloorTicks` exists only in comments, `_Gate0` L1185, `_TestGate0` L145).
4. `costFloorTicks` in integer cents: `tickValueCents = tick × bigPointValue × 100`, which must be within 1e-6 of an integer (else throw `mismatch:tickValue`); `costFloorTicks = ceil(roundTripCents × 10 / tickValueCents)` in `long` arithmetic, `(a + b − 1) / b`. ES: 1700 × 10 / 1250 → 14. MES: 250 × 10 / 125 = 20 exactly (no float path can give 21).
5. Costs: `p.cost` from `InstrumentConfig.costs` (cents). DECIDED: ES 1700, MES 250. Other roots: the method is DECIDED (Q3, IBKR fee per side ×2 + one tick slippage per side ×2), but the per-root IBKR fees are a **data input from Ettienne**; until given, cost = Unknown and cost users throw (S5).

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10. Tests (MC-free; new `tests/InstrumentTests` in `PLStudies.Tests.sln`, `CheckForOverflowUnderflow=true`, real sources linked via the `gen.sh` region pattern)

ID	Test
MI-T1	(Not a test.) `ClassOf` is the exhaustive host switch of §4.2, checked by the PLEditor compile in PR-M3. The MC-free core is tested on `EAssetClass` only.
MI-T2	Root resolver (§4.3): @ES, @ES#C, ESZ24, ESZ4, **ESZ2024 → ES, 'ES 12-24' → ES, ESZ24:XCME → ES, CLZ2026 → CL, @6E → 6E, 6EZ26 → 6E, EUR/USD → EURUSD, BRK.B → BRK (M1a; class-aware stock handling in the M2 resolver keeps BRK.B), Z4 → Z**, MESZ26 → MES, MESH27 → MES, **6E vs 6B differ**, @MNQ, GCQ26, ZNH27, **EUR/USD vs EUR/JPY differ**, BTCUSD, AAPL; M2: / removed for Spot FX only; M2 root source: MC root primary, parser fallback only on `(UnsupportedFields & eMTPA_SF_Root) != 0` (incl. -1) or empty root, `!SameRoot` → Warning + MC wins; **`_RootDump` fixtures (§4.4.3):** (@ES, ES, 0) → ES/mc, (@CL, CL, 0) → CL/mc, (@6E, 6E, 0) → 6E/mc, (SPY, SPY, 0) → SPY/mc, (@6E, "", 0) → 6E/ parsed, (@ES, ES, 1) → ES/parsed, (@ES, ES, -1) → ES/parsed, (@ES, ES, 512) → ES/mc; SymbolExpiry 2038-01-01 / 1900-01-01 → placeholder (calendar), 2026-12-18 → used; "ES" as a stock gets no preset; empty root rejected; `RootError("", "")` is an error (F1).
MI-T3	Preset cross-check: tick/bigPointValue mismatch throws `mismatch:tick` / `mismatch:bigPointValue`. MC-internal cross-check on the four `_RootDump` rows (§4.4.3) passes: 0.01 × 100, 1E-05 × 99999.999999999985 (within 1e-9), BPV 50/1000/125000/1 equal; perturbed rows (struct PriceScale 0.1 vs 100; BPV 50 vs 5) throw `mismatch:mcPriceScale` / `mismatch:mcBigPointValue` with the message naming both values.
MI-T4	Expiry rules vs CME calendar dates per root 2016–2027 (fixture CSV with the source URL); ES rows = `EventRules.RuleExpiryDate`.
MI-T5	Roll rules: the ES set reproduces `RollDatesDigest2000To2035 = "fdd63bd8f179a8d4"` (`_RegimeGate` L60).
MI-T6	`SeriesWindow` generalised: `tests/RollTests` unchanged; CL/GC monthly windows.
MI-T7	Trading day from sessions: CME Globex template (Sun 17:00 → Mon 16:00 CT, and the ET equivalent) gives `GlobexTradingDayEt` for every 1-min close over 2016–2024 incl. DST weeks; a two-session day (GLOBEX 17:00–15:15 + 15:30–16:30, the 9-entry list of `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`) shares one key; FX 17:00 ET template = ET+7h; 00:00 UTC crypto = UTC date; a day-only template keys by calendar date; a bar outside every session throws.
	Derived fields: `dayOpen`, `sessionEnd`, `hasOvernight`, `rthSplit` (day-only template → chart session; Globex template + ES → `EquityIndexLegacy` equal to `_CmeSessionClock` defaults field by field; other 24 h → null).

ID	Test
MI-T8	
MI-T9	Cost floor in integer cents: ES 14, **MES 20**, a non-integer tickValueCents throws.
MI-T10	Sizing: caps reduce q; the off-portfolio skip logs Info; floor off by default changes nothing; q never grows.
MI-T11	Exception text per reason code (§6.3), incl. `sessions:*`.
MI-T12	INSTR/CFG line formats pinned; culture-invariant (run under de-DE).
MI-T13	`profileHash` stable and outside every `configHash`.
MI-T14	Lint (in `tests/RepoLintTests`, PR-M1d) **scoped to MC enum types** (`ESymbolCategory`, `RequestTimeZone`, `EResolution`, `BuildVolumeOn`, `MTPA_MCSymbolCurrency`, `MTPA_MCSymbolCategories`, `MTPA_SymbolFields`, `DayOfWeek` from `SessionObject`): fails on `.ToString()` of those values feeding a comparison or a parse, on `Enum.Parse`/`TryParse` with those types, and on `System.Reflection` / `GetProperty(`. Exact allow-list (file + line pattern): `_CustomSignal` inputSnapshot reflection (L2254–2273), `_MarketBreadth.Strategy.CS` L7, `barZoneLabel` L279 (log), the L1883 `SeriesKeyWarning` text (log), `_TestPortMM` L90 (log). Repo-enum parses (`_EconomicCalendar` L690, `_StructuralTargetPatterns.Strategy` L305 / `.Indicator` L143, `_EventCalendar` L1307/L1318/L1322, `_PortMM` L233/L245) are out of scope. Runs after PR-M1b (L2462) and PR-M1e (probe) so it passes on that main.
MI-T15	ES bit-identity per PR: `_TestGate0` canary hash and the DVP D9a summary (**post-#88 baseline**) unchanged; ES fixtures in `tests/*` unchanged; requires **Ettienne's MC runs for every PR**, numbers pasted into the PR.
MI-T16	Session ends from the list (R14): the 9-entry GLOBEX list of `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt` with 1-min bars and `LastBarInSession` false at 15:15 (true only at 16:00, as dumped) gives session-end events at **15:15 and 16:00** each weekday and trading-day rollover after 16:00; a duplicate CalcBar at the same 16:00 time (`01M6HSDBFH2D6H21PMYXYKFJHV.txt`) resets once; a session-reset engine fixture (VPFUT/AVWAP/FLOW reset hook) sees exactly two resets per GLOBEX day and one per Default day; a list with no bar at its `EndTime` (data stops early) ends the session on the next out-of-session bar or on `LastBarInSession`.

11. PR list (each small; each targets `main`, opened after the previous one merged; no stacked PRs)

#	PR	Content	Behaviour change	Tests
—	**#88**	Land #86 + #87 (existing)	—	—
0	PR-M0	`_InstrDump.Strategy.CS` (§4.4, derives from `_CustomSignal`) + the `LogDirectory()` accessor in `_CustomSignal` (§4.4.1) + csproj entry + import note; `MCInstrDump_*.txt` output committed as fixtures after Ettienne's runs	none	existing suites
1	PR-M1a	F1: `_MtfMath.SymbolRoot` (§4.3 steps 1–2: leading digits + letter run, month drop, "/" removed; L109–119 on 6853398 before #90, L120–135 after #90), doc L28, `RootError` empty = error, roll inheritance and same-root checks	digit-led roots / FX only	MI-T2 (subset) in existing MtfMath/Dvp tests
2	PR-M1b	F2 (`isFuturesCategory` at L1069/L1866/L1882), F4 (tick throw), L2462 typed	FutureRolover, bad tick	existing suites
3	PR-M1c	F3a + F3b in `_ExitHalfRiskHalfRoll` only	legacy study only	—
4	PR-M1e	Retire `_InstrProbe` + `_InstrProbeCore` + `tests/InstrProbeTests` + csproj L521–522 + sln entry	none	suite count 23 → 22
5	PR-M1d	MI-T14 lint + allow-list	none	MI-T14
6	PR-M2	`_InstrumentProfile.Function.CS` core (enums, resolver, presets, `InstrumentConfig`, `CostModel`, exceptions, session derivations on structs), `<Compile Include>` + import order, `tests/InstrumentTests` + sln, rev-3 "superseded for C1–C9" banner (Q10)	none (not wired)	MI-T2–T9, T11, T13, T16 (core)
7	PR-M3	Wire `buildProfiles()` + `SessionReader` (StartCalc read + first-CalcBar re-check), `ClassOf` switch, INSTR line, cross-checks, `profileHash` on RUN	log only (+ mismatch throws)	MI-T12, T15
8	PR-M4	Trading day/day open/session end from sessions; remove L1888–1889, `autoTradingDay`, VPFUT Globex key/TOD 18:00; S1/S3/S7 rules; RTH split per N1; exchange map for ET only (NYMEX/COMEX unchanged, A3); session ends per R14 in the reader and every session-reset engine. PR-M0 output in §4.4.2	non-Globex templates and non-ES only	MI-T7, T8, T15, **T16**

#	PR	Content	Behaviour change	Tests
9	PR-M5	Expiry/roll from the profile: `_RollDetector`, `EventRules`, host quarterly loops (`_DvpEngineTest` L119–121, `_Test6` L100–102)	non-ES only	MI-T4–T6, T15
10	PR-M6a	G0: tick default removed + **Gate0Tests 35 constructions set tick**; root; S3	none for ES	Gate0Tests, T15
11	PR-M6b	FLOW: chunk boundary = `sessionEnd`	non-ES only	OrderFlowTests
12	PR-M6c	DAILY/REGIME `sessionsPerYear`	non-futures only	
13	PR-M6d	LEVEL/AVWAP tick defaults removed	none	AvwapTests, LevelTests
14	PR-M6e	RANGE/WEEKLY/PSTATS: S7 for `RthSessions`; CFG note; ringBars literal	none for ES	
15	PR-M6f	PORT cluster from profile	none	PortMM tests
16	PR-M6g	DVP roll table from profile	none for ES	DvpTests
17	PR-M6h	REGGATE: ES only (N2), explicit hash switches, S3	hash identical	RegimeGate tests, T15
18	PR-M7	Cost floor (opt-in, new) + costs in cents	none unless enabled	MI-T9, T10
19	PR-M8	Calendars for other roots, only when sourced (ES table unchanged)	non-ES only	calendar fixtures
later	PR-M9	Self-calibrating roll gap (Q4); ES stays 8 until it reproduces `tests/RollTests` and the m-event splices; `tools/m-event/out/roll_splices_2016_2024.csv` is read-only	none for ES	RollTests

Every PR lists the R1–R14 rules it touches, the VERIFY items it closes, its csproj/sln changes and the PowerLanguage Editor imports, and re-reads its line cites on current main.

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12. Questions

12.1 DECIDED

- **Q1 / Q2 Sessions and RTH windows: DECIDED by Ettienne 2026-11-02.** Verbatim: "The trading hours in questions 1 and 2 should be driven from Multicharts - it has session information you can get and use. This info is setup from the chart and just work with that. Each instrument also has a 'Default' setting which should be correct and used the traders in general." Applied as R12 and §4.5. Q2 (RTH windows for CL/GC/ZN/6E) is moot: no RTH windows in presets.
- **Q3 Costs for other roots: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): IBKR fees + one tick slippage, as ES/MES. **Data input outstanding:** IBKR per-side fees for NQ, MNQ, CL, GC, ZN, 6E; until given those roots have cost Unknown.
- **Q4 Roll-gap threshold: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): self-calibrating in PR-M9, only after it reproduces RollTests and the m-event splices; ES keeps 8 until then.
- **Q5 Roll rules for CL/GC/ZN/6E: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): rule-based table per §8.1, confirmed against the vendor's continuous splice; series detection fallback.
- **Q6 Retire `_InstrProbe`: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): PR-M1e.
- **Q7 Data-availability notes: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): `dataNotes`, logged on INSTR; FLOW bid/ask features undefined before the date (approved degraded mode D6, §6.4). **Data input outstanding:** dxFeed tick bid/ask start dates for @ES and @MES.
- **Q8 Non-USD: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): USD only; non-USD throws `currency`.
- **Q9 Stop floor: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): cost-derived, integer cents (§9.4), opt-in (PR-M7).
- **Q10 Supersede rev 3: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): banner in PR-M2.

12.2 N1–N6: DECIDED (Ettienne 2026-11-03 09:47 ET: "1 to 5 is fine, on 6 keep that dump in the same directory as the created standard log file")

- **N1 RTH split on a 24-hour chart: DECIDED as recommended (B).** ES/MES/NQ/MNQ on a 24-hour chart keep today's equity-index split in code (`rth src=EquityIndexLegacy`, ES bit-identical); other roots on a 24-hour chart throw S7 for RTH-dependent engines; on a day-only chart the chart session is the RTH. Option A (a second RTH-template stream) is not used because of MC-1439 stalls under Realtime-History Matching.
- **N2 A2-63 on MES: DECIDED as recommended.** REGGATE accepts ES only; MES would be a new trial with its own pre-registration.
- **N3 Overnight-only outputs without an overnight segment: DECIDED as recommended.** NaN/false with one Info line (U1–U2), not a throw.
- **N4 TOD half-day flag without a verified calendar: DECIDED as recommended.** Approved degraded mode D7 (§6.4).

- **N5 Asia/Europe/NY clock windows: DECIDED as recommended.** Kept as instrument-independent code defaults in ET (VPFUT L52–63, `_CmeSessionClock` L48–61), clipped to the chart's sessions; not presets.
- **N6 `_InstrDump` output: DECIDED.** Same directory as the standard MCSignal log, via the host's resolver (`_CustomSignal` `LogFileSingleton` L71–91, exposed through a new `LogDirectory()` accessor); file `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt` (§4.4.1).

12.3 Data inputs still pending (do not block PR-M0 to PR-M3)

- Q3: IBKR per-side fees for NQ, MNQ, CL, GC, ZN, 6E (until given: cost Unknown, S5).
- Q7: dxFeed tick bid/ask start dates for @ES and @MES.
- PR-M0 OPEN items (§4.4.2): Portfolio Trader second-symbol dump (@CL missing, ds2–10 absent); runs not yet made: @ZN, one FX pair, one crypto symbol.

Appendix A. Review responses (Coder review of rev 1, 37 items; each re-checked on `7dbdb1b`)

#	Item	Response	Where
1	Lint fails on main; needs allow-list, MC enums only	Accepted. Scoped to MC enums with an exact allow-list; remaining MC-enum string uses listed as findings	§3.3, MI-T14, PR-M1d
2	Removing `_InstrProbe` breaks InstrProbeTests	Accepted. Own PR with the suite, csproj L521–522 and sln; M1 does not depend on the M0 dump	PR-M1e
3	Removing G0 tick default breaks Gate0Tests	Accepted, with a correction: 35 new `Gate0Engine` constructions and **none** sets the engine tick (L168/184/191 are constructions); all 35 updated in PR-M6a	§7, PR-M6a
4	ringBars not ES-unchanged if derived	Accepted. Literals kept (2880; `_ProfileStats` 1440); they are bar counts	rows 55, 65; §7
5	§9.1 vs willArm	Accepted. §9 restates the code; floor is new opt-in	§9
6	Throw vs degrade conflicts	Accepted. One rule (R13), every case listed; rev-1 degrades withdrawn	R13, §6.4
7	Row 13 cite	Accepted: L1840–1842	row 13
8	Row 23 missing `MultiChartsProfile=0`	Accepted	row 23
9	Row 24 value window (09:30,16:00]	Accepted	row 24
10	Rows 26/28 VPFUT own table	Accepted	rows 26, 28
11	Counts	Accepted. One area per row, totals 70	§3.1
12	L2621 NYMEX/COMEX → Eastern	Accepted as a finding; map kept only for ET conversion; fixed in M4 if dump A3 shows Central	row 4, §4.4 A3, §4.5
13	Host quarterly tables	Accepted	row 64, PR-M5
14	L1888 Eastern	Accepted; removed by session-derived day key	row 63, §4.5
15	`_ProfileStats` 4-tick; NQ $2.50	Accepted	rows 66, 67
16	`SymbolRoot` doc	Accepted	row 68, PR-M1a
17	F1 wider	Accepted; MI-T2 adds 6E/6B and EUR/USD vs EUR/JPY	§3.2, MI-T2
18	F2 also L1069/L1866	Accepted	§3.2, PR-M1b
19	F3 description + second bug	Accepted	rows 60, 70; PR-M1c
20	F4	Accepted	PR-M1b
21	`Bars.Sessions` compiles	Accepted; StartCalc population, zone and per-series sessions CONFIRMED by the PR-M0 dump (rev 2.2); read at StartCalc, re-checked on the first bar	§4.1, §4.4 A2–A4, §4.5
22	`StrategyCurrencyCode` on plain signal	Accepted; moved to section B	§4.4 B2
23	Drop Description/PointValue/Resolution.Size from dump	Accepted	§4.4 A1
24	Dump3 hygiene	Accepted; cited as box-only run output; M0 commits typed output	header, §4.4
25	Rename to `bigPointValue`	Accepted	§5.1
26	Split M1	Accepted (a, b, c, d + e for the probe)	§11
27	Sequential PRs on main	Accepted; #88 first, no stacking	header, §11
28	M2 build plumbing	Accepted	PR-M2, §10 header
29	MI-T1 cannot be a test	Accepted; host exhaustive switch	§4.2, MI-T1
30	MI-T15 baseline + MC runs	Accepted; post-#88 baseline, Ettienne's runs per PR	MI-T15
31	Banner in M2	Accepted	PR-M2
32	R3 PriceAction preset	Accepted; preset clause dropped, "PriceAction unchanged" kept	R3

#	Item	Response	Where
33	R4 wording	Accepted: "no new [Input]s"	R4
34	REGGATE Q8 cross-ref; hash switch	Accepted; cross-ref fixed, MES as new trial raised as N2, explicit switches	§7, N2, PR-M6h
35	Cost floor exact integers	Accepted; integer cents, MES = 20 test	§9.4, MI-T9
36	FLOW boundary for FX	Changed: boundary = the chart's session end (no per-class rule); spot FX is already Unsupported for FLOW (rev-3 reason 11); 6E/FX futures use their own session end	§7, PR-M6b
37	Q4: ES stays 8; CSV read-only	Accepted	Q4, PR-M9

Rejected outright: none. **Partly changed:** item 3 (no Gate0 test sets the tick, so all 35 are updated) and item 36 (session end instead of a per-class rule).